Future of Time Series Anomaly Detection: From Historical Review to Unified Outlook

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Time series anomaly detection (TSAD), a major topic in data mining, focuses on recognizing patterns that dramatically depart from normal behavior in time series data, a work of crucial practical value. Driven by the rapid expansion in data volume and the rising complexity of application situations, recent years have witnessed the creation of various unique TSAD models, including diffusion models, State Space Models, and Large Language Models. However, existing reviews generally lack thorough coverage of these developing algorithmic designs and give little commentary on multivariate anomalies, cross-domain unified models, metric evaluation, and contemporary issues. To bridge this gap, this survey offers three key contributions: (1) We provide an updated anomaly categorization and algorithmic taxonomy that takes into account current research advances, outlining approaches from diverse paradigms, comparing them, assessing their developmental tendencies, and providing unique insights. (2) We conducted a thorough examination and analysis of existing evaluation methods, as well as an experiment to examine current measures. Furthermore, our open-source project website[1] contains a large variety of datasets and codebases, including algorithms, metrics, and analytical tools. (3) Through an open debate, we carefully explore the present status of algorithm evaluations and generalizable technologies, providing forward-looking talks on model interpretability, zero-shot learning, multi-modality/multi-domain learning, and foundational models. By summarizing current challenges and future opportunities, this review offers fresh perspectives for TSAD research and development.

Additional Key Words and Phrases: Time series anomaly detection, data mining, diffusion model, SSM, LLM

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1 Introduction

A time series is a sequence of data points indexed in chronological order, with extensive applications across diverse fields, from finance and healthcare to industrial monitoring and cybersecurity [5, 11, 20, 141]. Time series anomaly detection (TSAD), a critical branch of data analytics, aims to identify data points or segments that deviate significantly from normal patterns [164, 167, 185]. Its importance is paramount, as timely detection of anomalies can prevent catastrophic failures, such as identifying early signs of financial fraud, predicting equipment malfunctions, and detecting malicious network attacks. This capability is vital for optimizing system performance and ensuring the stable and secure operation of businesses and critical infrastructure [3, 4, 41, 114].

The field of TSAD has seen rapid evolution, from early statistical and classic machine learning methods to the powerful deep learning architectures of today. However, this rapid growth has also revealed notable limitations in existing research

[1] https://anonymous.4open.science/w/FTSAD-F1DF/

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Table 1. List of Abbreviations

Abbreviation	Expansion	Abbreviation	Expansion
TSAD	Time Series Anomaly Detection	Diff	Diffusion-based
SSM	State Space Model	CC	Channel-correlation
LLM	Large Language Model	FFN	Feed-Forward neural Network
RNN	Recurrent Neural Network	Rec	Reconstruction
GNN	Graph Neural Network	Fcst	Forecasting
AE	Autoencoder	Gen	Generative
VAE	Variational Autoencoder	Sim	Similarity
Prot	Prototype-based	Cls	Classification

surveys. As demonstrated by the detailed comparison in our Related Works (Sec. 2.1), previous surveys often focus on specific applications or older algorithms, leaving a significant gap in covering the latest algorithmic breakthroughs, such as Transformer [139, 164, 185], diffusion model [19, 55, 131, 162], State Space Model (SSM) [2, 49, 80, 159, 170], and Large Language Model (LLM). Furthermore, they often lack a refined taxonomy for current mainstream algorithms or do not revisit fundamental issues like anomaly classification and evaluation metrics. In the context of multivariate time series, the ambiguity in anomaly definitions further hinders research progress.

This paper presents a thorough survey to address these limitations. We aim to provide a new perspective on TSAD by introducing a refined classification for time series anomalies, proposing a structured taxonomy of mainstream algorithms, and delving into the design principles of state-of-the-art models. Importantly, we conduct open discussions on community-wide challenges and explore future trends in TSAD. The organizational framework of this paper is shown in Fig. 1.

In summary, our key contributions are:

(1) **A Refined Anomaly Taxonomy** (Sec. 3): We propose a new perspective on multivariate time series anomaly patterns, including the concept of channel-correlation anomalies, to provide a clearer framework for complex multivariate time series.
(2) **A Structured Algorithmic Taxonomy** (Sec. 4): We present a detailed classification of mainstream algorithms based on their unique structural characteristics, helping to reveal the core essence of their detection capabilities.
(3) **Thorough Review of Novel Algorithms** (Sec. B): We provide a detailed overview of the latest algorithms, with a special focus on Diffusion Models, State Space Models, Prototype-based Models, and LLMs, which are not consistently covered in previous surveys.
(4) **Review and Analysis of Evaluation Metrics** (Secs. 6 & 7): We extensively review and analyze current evaluation metrics, highlighting their problems and offering practical suggestions for their selection. To facilitate future research and practice, we have compiled and publicly released implementations of mainstream algorithms, commonly used benchmark datasets, and evaluation metrics.
(5) **Open Discussions & Future Outlook:** (Secs. 7 & 8): We spark discussions on critical open issues and future research challenges, such as zero-shot learning, multi-modality, and foundational models, to provide new insights for the community.

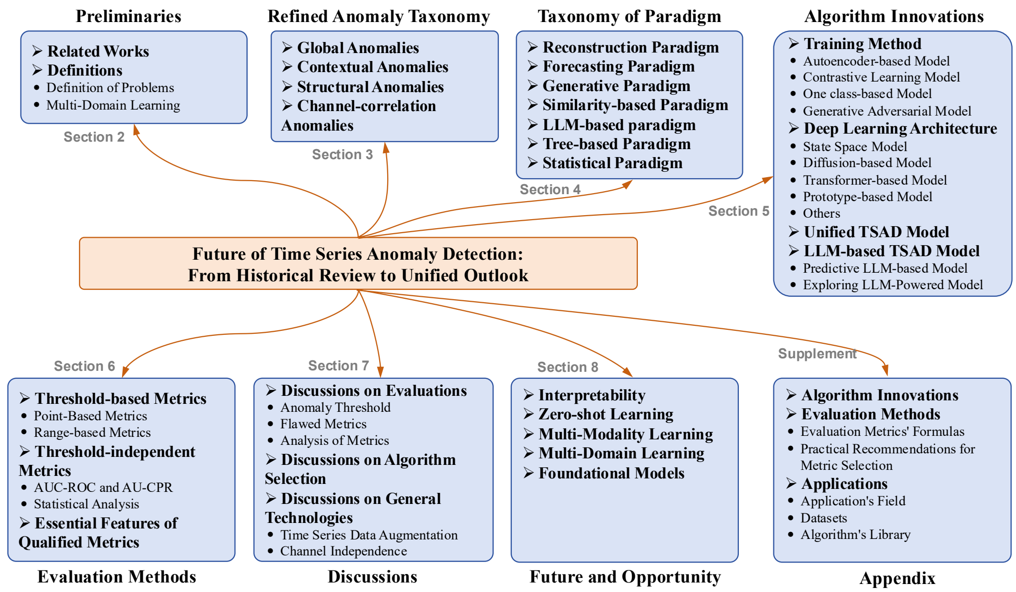


Fig. 1. The organizational framework of this paper.

In the remainder of the review, Section 2 first surveys related work and then defines core concepts, situating our survey within the existing literature before introducing fundamentals. Section 3 presents our structured taxonomy of algorithms. Section 5 covers novel TSAD algorithms, while Section 6 discusses their evaluation methods. Finally, Sections 7 and 8 are dedicated to open discussions and future directions. The common acronyms used throughout the paper are listed in Table 1.

Additionally, all resources mentioned in this survey, including supplemental materials, code, and data, are available on our project website at https://anonymous.4open.science/w/FTSAD-F1DF/.

2 Preliminaries

This section first introduces related work, positioning our survey within the landscape of existing TSAD literature. It then defines the key terminology used throughout this survey to facilitate understanding of Secs. 3 & 4.

2.1 Related Works

This survey categorizes the evolution of TSAD algorithms into four distinct phases, providing a simplified framework based on their primary characteristics to help readers understand the community's development.

The **first phase** focused on **classical machine learning algorithms**. During this period, researchers applied general-purpose machine learning algorithms for anomaly detection to time series data, or they proposed improvements tailored to time series [85, 88, 115, 145]. While simple and interpretable, these methods often struggle with the complexity and high dimensionality of modern time series data. This is because they assume the data have a linear or low-rank structure, whereas the dependencies in modern high-dimensional time series are more complicated, nonlinear, and dynamic.

The **second phase** saw the emergence of **deep learning models with a small number of parameters for TSAD**. These models typically have fewer parameters and primarily include four architectural types: Recurrent Neural Networks (RNNs) [100, 130, 160, 187], Temporal Convolutional Networks (TCNs) [1, 9, 56, 110, 114], Graph Neural Networks (GNNs) [31, 47, 57, 163], and Prototype-based Models (Prot) [128, 179, 184]. These algorithms are characterized by their small parameter count, limited ability to capture long-range dependencies, and fast inference speeds. Previous surveys have paid less attention to Prot algorithms, so this survey synthesizes and introduces this category of algorithms.

The **third phase** centers on **large-parameter deep learning models for TSAD**. In recent years, this phase has witnessed the emergence of several innovative backbone architectures that represent significant paradigm shifts. Compared to the previous phase, the TSAD algorithms in this phase utilize more advanced network architectures that can capture much longer-range temporal dependencies, albeit with greater computational complexity and a larger number of parameters. Transformer-based models leverage self-attention to capture long-range dependencies [139, 164], while Diffusion-based Models (Diff) model the data distribution to identify low-probability instances [55, 112]. State Space Models (SSMs), particularly Mamba-based model, have shown promise in efficiently modeling long sequences [2, 80]. Previous surveys have only focused on Transformer-based models and have not given sufficient attention to diffusion models and SSMs. To address this gap, we consolidate and review these two algorithm categories.

Finally, the **fourth phase** explores **future-oriented deep TSAD algorithms**. This phase includes two types of algorithms: Unified TSAD models and LLM-based TSAD models. These are considered future-oriented because they aim to tackle more complex, critical, and challenging tasks in time series anomaly detection. We will discuss these in detail in Secs. 7 and 8. These models need to enhance their interpretability and zero-shot inference capabilities, solve challenges in multi-modal and multi-domain learning, and explore the potential for future TSAD Foundation Models. However, existing surveys rarely focus on LLM-based models and lack in-depth discussions on these future challenges and opportunities.

Table 2 compares our work with related surveys, ordered by publication date, to position our contributions within the existing literature. As the table illustrates, our survey is distinguished from previous works in several key aspects:

- **More Extensive Coverage:** Existing surveys have not simultaneously covered a range of emerging deep learning algorithms, such as Prot, SSM, Diff, and LLM (as indicated in the "Novel Algorithms" column). Furthermore, the concept of a Unified TSAD Model has not been explored in prior works. Our survey provides a more extensive review and in-depth discussion of these cutting-edge algorithms and their applications.
- **A New Taxonomy of Paradigm:** With the continuous emergence of new algorithms, we introduce a novel framework for classifying paradigms. This framework re-categorizes existing algorithms and summarizes the core characteristics of each paradigm. In Appendix A, we analyze the development trends of different paradigms.
- **A More Refined Anomaly Taxonomy:** Although previous surveys have proposed taxonomies for time series anomalies [14, 22, 38, 119], they are generally more suitable for univariate time series. For multivariate time series, anomaly detection must account for inter-channel relationships [8, 185, 186]. Therefore, building upon prior work, we propose a more refined taxonomy for anomalies that is better suited for the complexities of multivariate time series.

2.2 Definitions

This subsection provides key definitions in TSAD. These definitions are crucial for understanding the anomaly taxonomy presented in Sec. 3 and the paradigm taxonomy discussed in Sec. 4.

2.2.1 Definition of Problems. **Time series data** is a sequence of data points arranged in chronological order. Formally, a time series can be represented as $\boldsymbol{t} = [t_1, t_2, \cdots, t_N] \in \mathbb{R}^{N \times C}$, where t_i denotes the data point at time i, and N signifies the total number of observations. At each time point, it consists of a set of multivariate data composed of different channel features (e.g., C channels, variables), meaning $t_i \in \mathbb{R}^C$. The time series data for the c-th channel can be denoted as $\boldsymbol{t}_{:,c} = [t_{1,c}, t_{2,c}, \cdots, t_{N,c}] \in \mathbb{R}^N$. A time series with different channel features is referred to as **multivariate time series**. Specifically, when $C = 1$, indicating only one channel feature, it is referred to as **univariate time series**. The time series data is characterized by its continuity and the inherent correlation between consecutive data points.

For each time point, a label y_i is provided to indicate whether that moment is anomalous. When $y_i = 1$, it signifies an anomaly, while $y_i = 0$ indicates the time point is normal. Therefore, for $\boldsymbol{T}$, there is a corresponding $\boldsymbol{y} = [y_1, y_2, \cdots, y_N]$. Typically, the number of anomalous time points is much fewer than the number of normal time points. Given a multivariate time series $\boldsymbol{T}$, we aim for algorithms to accurately predict the label for each time point as much as possible.

Anomaly Length: If $y_i = 1 \wedge y_{i-1} = 0 \wedge y_{i+1} = 0$, it is said that a point anomaly occurs at time point i. If for a period of time from time point i to j, $y_{i:j} = 1$ is satisfied, that is, this interval can be called a segment/sequence anomaly [167]. Meanwhile, $L = j - i + 1$ is called the duration of the anomaly or the anomaly length. Anomalies of different durations tend to correspond to anomalies of different severity levels [61].

2.2.2 Multi-Domain Learning. **Multi-domain Learning**, in the context of time series analysis, refers to the process of learning from and across multiple domains of temporal data [96, 122]. Specifically, a **multi-domain temporal dataset** can be denoted as $\mathcal{D} = \{\mathcal{T}, \mathcal{Y}, \mathcal{S}\}$, where $\mathcal{T}$, $\mathcal{Y}$, and $\mathcal{S}$ represent the sets of multi-domain **time series data**, **labels**, and **domains**, respectively. Here, $\mathcal{T} = \{\boldsymbol{t}^1, \boldsymbol{t}^2, \ldots, \boldsymbol{t}^{N_d}\}$ denotes a **multi-domain time series dataset** comprising temporal data from N_d distinct domains. Each $\boldsymbol{t}^i$ can be expanded to represent the observed variables of a specific domain, expressed as $\boldsymbol{t}_{:,c} = [t_{1,c}, t_{2,c}, \cdots, t_{N,c}]$. Similarly, the **label set** across multiple domains is represented as $\mathcal{Y} = \{\boldsymbol{y}_1, \boldsymbol{y}_2, \ldots, \boldsymbol{y}_{N_d}\}$, where each $\boldsymbol{y}_i$ is given by $\boldsymbol{y}_i = \{y_1, y_2, \ldots, y_N\} \in \mathbb{R}^N$. Different domains vary in temporal data lengths N_i and the number of channels C_i, such that for any two domains i and j within the range $[1, N_d]$, $(N_i \neq N_j) \vee (C_i \neq C_j)$. This variability necessitates that the model be capable of learning from datasets with diverse temporal lengths and channel counts. The goal of multi-domain learning is to effectively leverage the shared knowledge across different domains while accounting for their unique characteristics, thereby enhancing the model's ability to generalize and perform accurate predictions in a multi-domain setting.

Unified Time Series Anomaly Detection Models aim to detect anomalies across diverse temporal domains while addressing the inherent heterogeneity in multi-domain settings [121]. Formally, given the multi-domain dataset $\mathcal{D}$, a unified model $\mathcal{M}$ is defined as a parameterized function that maps temporal inputs $\mathcal{T}$ to anomaly scores $\mathcal{A}$ and domain-aware representations $\mathcal{Z}$, i.e., $\mathcal{M}(\mathcal{T}; \Theta, \Phi) \rightarrow (\mathcal{A}, \mathcal{Z})$, where Θ denotes **domain-shared parameters** capturing cross-domain temporal patterns, and $\Phi = \{\phi_1, \phi_2, \ldots, \phi_{N_d}\}$ represents **domain-specific parameters** to adapt to unique characteristics of each domain. Here, domain-specific parameters are optional, depending on whether the model knows what the new or unseen domain is.

Such models should be **a shared model** and achieve **two-level generalization**: *One shared model:* In different domains $\mathcal{D}$, the unified model $\mathcal{M}$ should simultaneously share the multi-domain parameters Θ [36]. *Intra-domain*: Detecting anomalies under temporal distribution shifts within a domain. *Cross-domain*: Transferring detection capabilities to unseen domains with minimal adaptation. The unified model bridges the gap between heterogeneous temporal domains while maintaining computational efficiency through parameter sharing, making it scalable for real-world multi-domain monitoring systems [36].

Table 2. Overview of Time Series Anomaly Detection Surveys.

Citation	Application	Taxonomy of Paradigm	Anomalies Taxonomy	Novel Algorithms		Library	Datasets	Metrics	Discussion		
				Prot/SSM/Diff	LLM				Interpretability	Multi-Modality	Multi-Domain
Landauer et al. [77] (2018)	✓							✓			
Zhao et al. [181] (2019)	✓						✓				
Cook et al. [22] (2020)	✓		✓								
Braei and Wagner [15] (2020)							✓	✓			
Blázquez-García et al. [14] (2020)			✓			✓					
Choi et al. [20] (2021)	✓	✓					✓				
Shaukat et al. [119] (2021)			✓								
Lai et al. [76] (2021)			✓			✓	✓				
Garg et al. [38] (2021)		✓				✓	✓	✓			
Zhang et al. [175] (2021)			✓								
Schmidl et al. [117] (2022)		✓	✓				✓	✓			
Wu and Keogh [153] (2022)							✓				
Alimohammadi et al. [5] (2022)		✓	✓								
Kim et al. [73] (2022)								✓			
Belay et al. [11] (2023)		✓	✓						✓		
Jin et al. [70] (2023)	✓	✓				✓			✓		✓
Kim et al. [71] (2023)	✓										
Wen et al. [148] (2023)											✓
Li and Jung [81] (2023)	✓		✓								
Yan et al. [161] (2023)	✓	✓	✓								
Sun et al. [133] (2024)								✓			
Ye et al. [165] (2024)	✓				✓				✓	✓	✓
Su et al. [129] (2024)	✓				✓		✓	✓	✓		
Zamanzadeh Darban et al. [167] (2024)	✓	✓	✓	✓				✓	✓		
Ours	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

3 Refined Anomaly Taxonomy

An anomaly refers to a deviation from the general data distribution, appearing as either an individual observation (point) or a series of observations (subsequence) that significantly deviates from the norm [76].

Historically, the literature [15, 76, 81, 175] has often classified anomalies as global, contextual, and structural. However, this classification method is primarily suited for univariate time series. In this section, we first introduce the three classic types of anomalies. Subsequently, in Sec. 3.4, we introduce the concept of channel-correlation anomalies to describe more complex anomalies in multivariate time series.

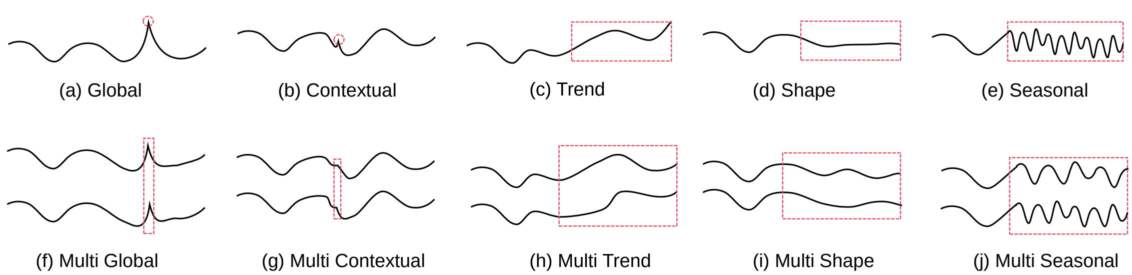


Fig. 2. Illustration of different types of anomalies. The red box indicates the area of anomaly.

3.1 Global Anomalies

Global anomalies are outliers within a time series, often appearing as sudden "spikes" indicating rare or significant deviations [15, 167] (Fig. 2(a)). They can be defined by:

$$\mathcal{F}_{ga} = |t_i^c - \hat{t}_i^c| > \delta_{ga}. \tag{1}$$

Here, $\mathcal{F}_{\text{type}}$ (e.g., $\mathcal{F}_{ga}$) denotes the anomaly score, and δ_{type} (e.g., δ_{ga}) is the corresponding threshold. $\hat{t}_i^c$ represents the predicted value for channel c at time point i, and t_i^c is the observed value. If the absolute difference between the observed and predicted values at a specific time point i for channel c exceeds the threshold δ_{ga}, it is identified as a global anomaly.

3.2 Contextual Anomalies

Contextual anomalies are data points that appear anomalous in one context but are normal in others [15, 151, 167]. Their identification depends on the contextual information associated with the data (Fig. 2(b)). This can be mathematically represented as:

$$\mathcal{F}_{ca} = |\sigma(\boldsymbol{t}_{(i-m+1:i),c}) - \sigma(\hat{\boldsymbol{t}}_{(i-m+1:i),c})| > \delta_{ca}. \tag{2}$$

In this equation, $\boldsymbol{t}_{(i-m+1:i),c}$ signifies the observed data for channel c within a time window of length m preceding and including time point i, while $\sigma(\cdot)$ indicates a function measuring the contextual deviation (e.g., standard deviation or another contextual feature) of that window.

3.3 Structural Anomalies

Trend anomalies arise from unexpected drastic changes in the trend component of a time series, signaling a disruption in the long-term progression [15] (Fig. 2(c)).

Shape anomalies occur when the local pattern of a time series differs significantly from the overall sequence shape or expected pattern (Fig. 2(d)).

Seasonal anomalies refer to deviations from expected seasonal patterns in a time series, where data points, though not extreme in magnitude, show significant differences from these anticipated cyclical variations (Fig. 2(e)).

These anomalies can be unified under the following general formula:

$$\mathcal{F}_{sa} = \text{StrucFunc}_{\text{type}}(\boldsymbol{t}_{(i-m+1:i),c},\ \hat{\boldsymbol{t}}_{(i-m+1:i),c}) > \delta_{sa} \tag{3}$$

Here, $\text{StrucFunc}_{\text{type}}$ represents various structural evaluation functions (e.g., $\text{StrucFunc}_{\text{seasonal}}$, $\text{StrucFunc}_{\text{trend}}$, $\text{StrucFunc}_{\text{shape}}$). Typically, anomaly scores for structural anomalies are derived by analyzing the decomposed components (e.g., periodic and trend) of the time series.

3.4 Channel-correlation Anomalies

Previous work has proposed increasingly sophisticated anomaly definitions [15, 76, 81, 175], yet it rarely clarifies the boundary between multivariate and univariate settings, nor does it quantify how inter-channel relationships contribute to the final anomaly score. Many studies [8, 76, 164] adopt the channel-independence assumption (see Sec. 7.3.2), treating each channel as an isolated univariate series. This simplification is valid only when cross-channel dependencies are negligible [167].

Building on the notion of correlation anomalies introduced by Wenig et al. [151], we present a unified view: Channel-correlation (CC) anomalies arise from simultaneous abnormal behaviors across multiple channels, manifesting as anomalous joint distributions rather than extreme values in any single variable.

Figs. 2(f-j) illustrate five types of channel-correlation anomalies: CC-global, CC-contextual, CC-trend, CC-shape, and CC-seasonal. For example, in Fig. 2(g), even though the peaks in the two channels are higher than their usual levels, the fact that they happen at the same time is what makes this point unusual. In Fig. 2(i), the shapes of both channels largely adhere to their prior patterns, causing traditional shape-detection rules to classify the point as normal. However, under the CC-shape criterion, the subtle, simultaneous shape distortions observed across the two coupled channels collectively constitute an anomaly.

We formalize the anomaly score for a single channel c as $\mathcal{F}^c_{\text{type}}$. Channel-correlation anomalies can be defined as follows:

$$\begin{gathered}\mathcal{F}_{\text{cca}} = \sum_{c=1}^{C} \mathcal{F}^c_{\text{type}} > \sigma_{\text{cca}} \\ \text{s.t. } \forall \text{type} \in \{\text{ga}, \text{ca}, \text{sa}\}, \forall c, \quad \mathcal{F}^c_{\text{type}} > \sigma_{\text{type}}.\end{gathered} \tag{4}$$

This formulation detects a channel-correlation anomaly when every single-channel score $\mathcal{F}^c_{\text{type}}$ exceeds its individual threshold σ_{type}, and their aggregate $\sum_c \mathcal{F}^c_{\text{type}}$ simultaneously surpasses the joint threshold σ_{cca}. This simultaneous violation across all channels identifies anomalous behavior specific to multivariate time series.

In practice, pinpointing which underlying anomaly type (global, contextual, or structural) drives a channel-correlation anomaly is often impractical. A simpler and widely adopted criterion therefore assesses each channel's overall contribution to the joint score, ignoring the concrete anomaly sub-type within each channel:

$$\begin{gathered}\mathcal{F}_{\text{cca}} = \sum_{c=1}^{C} \alpha_c \, \mathcal{F}^c \; > \; \sigma_{\text{cca}}, \\ \text{with } \forall c, \; \mathcal{F}^c > \sigma_c,\end{gathered} \tag{5}$$

where $\alpha = [\alpha_1, \ldots, \alpha_C]$ are predefined channel weights and $\mathcal{F}^c$ is the overall anomaly score for channel c. This formulation quantifies the causal influence of each channel on the final anomaly. Importantly, α_c allows irrelevant channels to be excluded by setting their weights to zero when only a subset of channels actually drives the anomaly.

Equations (4) and (5) specify how to recognize channel-correlation anomalies, not how they are generated. Because real-world anomalies can stem from diverse and intricate inter-channel dependencies, we adopt a concise detection rule that focuses on observable synchrony rather than modeling the underlying causes.

Figs. 2(f-j) illustrate five types of channel-correlation anomalies: CC-global, CC-contextual, CC-trend, CC-shape, and CC-seasonal. In Fig. 2(g), although the peaks in the two channels individually exceed their contextual thresholds, their synchronous occurrence is what identifies this point as anomalous. In Fig. 2(i), the shapes of both channels largely adhere to their prior patterns, causing traditional shape-detection rules to classify the point as normal. However, under the CC-shape criterion, the subtle, simultaneous shape distortions observed across the two coupled channels collectively constitute an anomaly.

4 Taxonomy of Paradigm

In this section, we establish a paradigmatic classification of existing algorithms based on their anomaly detection capabilities and contributions, and introduce their basic processes and formal descriptions.

Figure 3 illustrates the underlying principles behind the taxonomy of paradigms. Building on this, Fig. 4 maps individual algorithms to these paradigms based on their training regimes and model architectures. Finally, Table S7 summarizes, for each paradigm, every algorithmic category covered in this paper together with its key references.

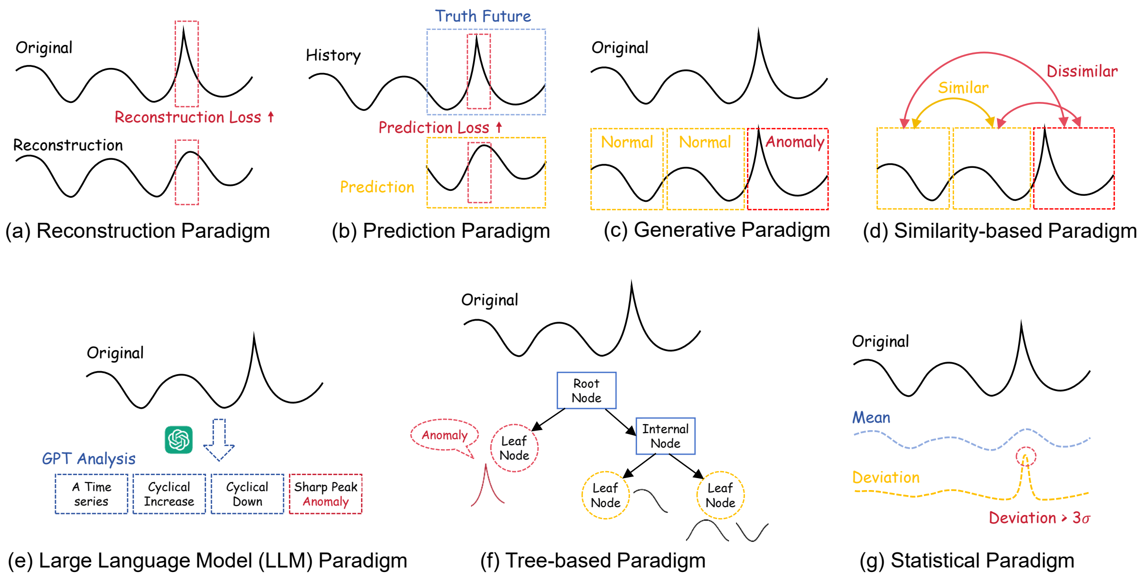


Fig. 3. Diagram of different paradigms of TSAD.

4.1 Reconstruction Paradigm

The reconstruction paradigm (Fig. 3(a)) functions as an anomaly detection framework that leverages the intrinsic structural properties of time series, with a particular focus on their temporal continuity (i.e., the smooth transition between consecutive data points over time) and cross-dimensional correlation (including both intra-series dependencies across time steps and inter-channel relationships in multivariate scenarios). The core concept suggests that the inherent data structure of time series exhibits a specific trait: successful reconstruction from neighboring points indicates normality, while a substantial reconstruction error implies potential abnormality [164, 167, 185].

In a time series $\boldsymbol{t} = [t_1, t_2, \cdots, t_N]$ where each $t_i \in \mathbb{R}^C$, the reconstruction paradigm is defined as $f_{rec} = \{f_{enc}, f_{dec}\}$. Here, f_{enc} serves as the encoder converting a continuous data segment into fixed-size features denoted as $\boldsymbol{z}$, and f_{dec} acts as the decoder reconstructing the data from this encoded features. Together, they constitute an autoencoder framework designed for efficient representation learning. The autoencoder is composed of two main parts: an encoder $f_{enc} : \mathbb{R}^{m \times C} \rightarrow \mathbb{R}^d$ and a decoder $f_{dec} : \mathbb{R}^d \rightarrow \mathbb{R}^{m \times C}$. Here, m represents the time window size, and d indicates the dimension of the encoded features. This autoencoder processes a time window $\boldsymbol{t}_{(i-m+1):i} = [t_{i-m+1}, t_{i-m}, \cdots, t_i]$. Initially, it encodes the data into features as follows:

$$\boldsymbol{z} = f_{enc}(\boldsymbol{t}_{(i-m+1):i}), \tag{6}$$

and subsequently reconstructs it from the encoded features with:

$$\hat{\boldsymbol{t}}_{(i-m+1):i} = f_{dec}(\boldsymbol{z}). \tag{7}$$

The reconstruction error, often measured using mean squared error (MSE) [7, 74], serves as the anomaly assessment score:

$$\mathcal{F}_i^{\text{rec}} = \frac{1}{m} \sum_{k=i}^{i-m+1} \|t_k - \hat{t}_k\|_2^2. \tag{8}$$

If this anomaly score exceeds a threshold δ (*i.e.*, $\mathcal{F}_i > \delta$), the time point i is flagged as potentially anomalous.

4.2 Forecasting Paradigm

The forecasting paradigm (Fig. 3(b)) is an anomaly detection approach that focuses on forecasting future values of time series rather than reconstructing past data points [36, 56]. In other literatures, such as [37, 167, 175], this type of model is also referred to as the **Predictability model** or **Prediction-Based model**. Therefore, in the absence of ambiguity, the words predictability or prediction are not distinguished from forecasting. Anomalies are identified by comparing predicted values with actual observations; significant discrepancies indicate anomalies. This method is based on the understanding that anomalous points adversely affect future predictions and reduce the accuracy of forecasting models. Monitoring prediction errors enables us to detect points that significantly disrupt the model's predictive performance.

Assuming a time series forecasting model $f_{pred} : \mathbb{R}^{C\times m} \rightarrow \mathbb{R}^{C\times p}$, where m is the size of the prediction window, the model forecasts the values of the next p time points based on the window $\boldsymbol{t}_{(i-m+1):i} = [t_{i-m+1}, t_{i-m+2}, \cdots, t_i]$:

$$\hat{t}_{(i+1):(i+p-1)} = f_{pred}(\boldsymbol{t}_{(i-m+1):i}). \tag{9}$$

The prediction error, calculated as the anomaly assessment score using mean squared error (MSE):

$$\mathcal{F}_i^{\text{pred}} = \frac{1}{p} \sum_{k=i}^{i+p-1} \|t_k - \hat{t}_k\|_2^2. \tag{10}$$

If this anomaly score exceeds a specified threshold, the time point is flagged as anomalous.

Notably, [92, 144, 164, 186] have experimentally analyzed the anomaly detection capabilities of various prediction models, revealing that models trained for prediction can be directly employed for anomaly detection, provided that different layers of the model learn consistent feature representations.

4.3 Generative Paradigm

The generative paradigm (Fig. 3(c)) detects anomalies by learning how normal data are produced. It first fits a generative model to the training data, then flags any observation that receives unusually low likelihood under this model or whose synthetic neighbours look markedly different ([33, 39, 93, 169]). In other words, if the model finds it hard to generate a point or assigns it negligible probability, that point is treated as anomalous. Although many implementations use reconstruction as an intermediate step, the key distinction from reconstruction paradigms is the ability to *generate* data: reconstruction methods can only *discriminate* whether an input is abnormal, whereas a generative method can also sample new anomalies from the learned distribution.

Assuming a generative model $G : \mathbb{R}^d \rightarrow \mathbb{R}^{C\times N}$, where d represents the latent space dimension and N is the length of the generated time series, the model generates time series data $\hat{\boldsymbol{t}}$ from a latent variable $\boldsymbol{z}$.

To begin, we optimize the parameters θ of the generative model through methods like maximum likelihood estimation, ensuring that $G(\boldsymbol{z}; \theta)$ closely aligns with the real data distribution. This is achieved by minimizing the loss function:

$$\mathcal{L}(\theta) = \frac{1}{N} \sum_{i=1}^{N} \|\boldsymbol{t} - G(\boldsymbol{z}; \theta)\|_2^2. \tag{11}$$

Next, for each observed data point t_i, we sample a latent variable $\boldsymbol{z}$ from the prior distribution $p(\boldsymbol{z})$. The generative model G is then employed to generate the data point $\hat{t}_i = G(\boldsymbol{z}; \theta)$ based on the sampled latent variable.

To compute the anomaly score, we utilize a metric $\mathcal{D}$ to quantify the difference between the observed data point t_i and the generated data point $\hat{t}_i$:

$$\mathcal{F}_i^{\text{gen}} = \mathcal{D}(t_i, \hat{t}_i). \tag{12}$$

Common choices for the metric $\mathcal{D}$ include MSE or Kullback-Leibler (KL) divergence [23], though other methods can also be used, such as employing a separate neural network to evaluate distinctiveness, similar to approaches used in Generative Adversarial Network (GAN) [44].

4.4 Similarity-based Paradigm

The similarity-based paradigm (Fig. 3(d)) is a type of anomaly detection model that determines the anomaly of data points by computing similarity measures between the data points.

Formally, given a time series data $\boldsymbol{t} = [t_1, t_2, \cdots, t_N]$, in the naive case, we treat each time point as a data sample. Subsequently, a similarity measure function $\text{sim}: \mathbb{R}^c \times \mathbb{R}^c \to \mathbb{R}$ is defined to measure the similarity between two data points. Similarity functions are created using specialized modeling approaches or feature engineering. And based on the underlying approach, these functions can be divided into three subcategories. This taxonomy attempts to distinguish the different design philosophies and key ideas of the various methods. It is crucial to note, however, that all of these functions fall within the wider Similarity-based Paradigm.

(1) **Distance-based Function**: *Anomalies are data points that lie at a significant distance from the majority of other points.* Using metrics such as Euclidean distance or Dynamic Time Warping (DTW), this function identifies anomalies by comparing distances to a predefined threshold.

(2) **Density-based Function**: *Normal data points are in high-density regions, whereas anomalies are in low-density regions.* This function calculates local densities using algorithms like LOF to determine anomaly scores based on the density of a point relative to its neighbors.

(3) **Classification-based Function**: *Anomaly detection can be framed as a binary classification problem where normal data lies at the center of a high-dimensional sphere and anomalies lie outside the sphere.*

4.5 LLM-based paradigm

The LLM-based paradigm (Fig. 3(e)) is an anomaly detection approach that employs large models as anomaly detectors or integral components of anomaly detection systems [13, 28, 75, 87, 91, 106, 188]. Leveraging the autoregressive capabilities of LLMs and their strong performance in sequence tasks like natural language processing, these models can be adapted for TSAD in various ways, which can be categorized into two main strategies:

(1) Injecting External Knowledge: For targeted TSAD tasks, external knowledge can be incorporated into LLM through prompt engineering or by utilizing knowledge repositories such as Retrieval Augmented Generation (RAG). By embedding external knowledge, pre-trained LLM can function as anomaly detectors for zero-shot TSAD [13, 25]. This process assumes the existence of a model LLM that processes input text to generate specific text-based responses, creating a mapping: $\mathsf{LLM}: \text{text} \to \text{text}$. Given that time series data is usually numerical, a method is necessary to convert this data into textual descriptions. For instance, a time series like $[1, 3, 5, 10]$ could be described as *The time series rises from value 1 to value 10, passing through values 3 and 5, showing an overall increasing trend.* This establishes a knowledge injection mapping $\mathsf{Inject}: \mathsf{timeseries} \to \text{text}$. Subsequently, researchers need to design specific system

prompts to guide the LLM. For example, the prompt could be *You are a general-purpose assistant in the field of time series. Please analyze the given time series text to identify any abnormal periods and explore the potential causes of these abnormalities.* By integrating such system prompts with the LLM's inherent text generation capabilities, the LLM can pinpoint abnormal segments in the time series and describe the abnormalities in natural language [13].

(2) Fine-tuning: LLM involves freezing the self-attention layers, FFN, or both, which preserve the extensive pre-trained knowledge of the models [36, 87, 188]. This process enables the adjustment of positional embeddings and layer normalization layers within LLMs, or selectively fine-tuning the frozen parameters, as well as introducing parallel side-adapters. These adaptations help LLMs understand the relationship between time series and text, facilitating the learning of temporal features and enabling few-shot TSAD. Various efficient fine-tuning techniques, including Adapters, Prompt Tuning, Prefix-Tuning, and Low-Rank Adaptation (LoRA), can be utilized in the fine-tuning implementation.

In this latter mode, the goal is to transform the modal data that the existing LLM model can handle from text to time series, resulting in a new mapping: $\mathsf{LLM}' : \mathsf{timeseries} \rightarrow \mathsf{timeseries}$. The motivation behind this mode is based on the belief that LLMs have already acquired general knowledge, including knowledge of time series data, and thus possess text generation capabilities. However, since LLMs are trained on natural language, their representation of time series data is incomplete, necessitating appropriate fine-tuning of LLM to obtain a more suitable LLM' for time series data. Formally, assuming the original LLM has trainable parameters θ_{LLM}, a subset of parameters θ'_{LLM} can be selected from it, or new parameters θ_{adp} for additional side-adapters can be introduced. By fine-tuning the combined set of parameters $\{\theta'_{LLM}, \theta_{adp}\}$, the LLM can be adapted to time series data to detect anomalies. In a straightforward approach, the LLM can be transformed into a prediction-based model, detecting anomalies based on prediction errors.

4.6 Tree-based Paradigm

The tree-based paradigm (Fig. 3(f)) employs models with tree structures, utilizing various algorithms such as Isolation Forest and Decision Trees to construct these trees. The creation of these structures is based on the similarity measures between data points. The Isolation Forest model, for instance, builds multiple random partition trees by recursively selecting random cut points and partitions within the dataset. For each data point, the average path length across all trees is calculated and serves as the anomaly score. This approach enables effective detection of anomalies in time series data.

4.7 Statistical Paradigm

The statistical paradigm (Fig. 3(g)) utilizes statistical methods to assess anomalies in data points through analysis and modeling of observed data. This typically involves analyzing the periodicity and trend information in time series data. Metrics such as the mean and standard deviation are often calculated to evaluate anomalies. Common techniques include box plots and z-scores, among others.

5 Algorithm Innovations

To highlight the key points of current research and our contributions, ensure the orthogonality of the article's narrative and algorithm classification, and minimize confusion for readers, we divide the content into five parts. Fig. 4 illustrates the relationships among the taxonomy of paradigms, deep learning training methods, deep learning architectures, and classical machine learning approaches, aiming to facilitate the understanding of the narrative presented in this section. It is important to note that different paradigms necessitate specific training methods and are associated with particular algorithmic architectures. A more in-depth analysis of these intricate relationships is provided in Sec. 7.2.

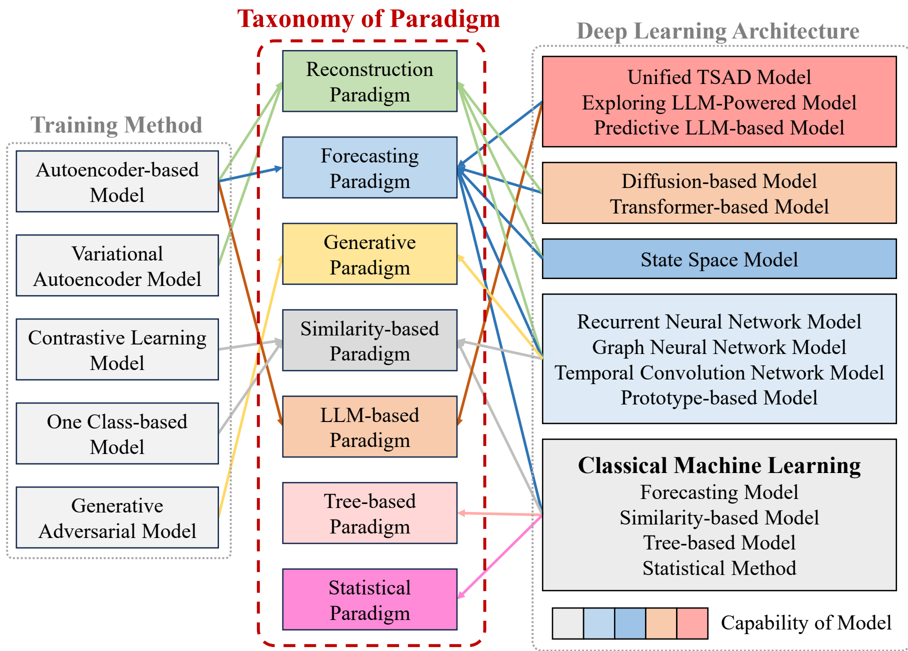


Fig. 4. The corresponding relationship among training methods, different architectures (including deep learning and classical machine learning), and the taxonomy of paradigm.

Table S7 summarizes all the algorithmic categories reviewed in this paper together with their key references.

(1) **Training Method** (Sec. 5.1): Starting from the original design intentions of past deep learning models, introduce the current mainstream model training/utilization methods. At the end of this subsection, we conduct a comparative analysis of its sub-types, highlighting the strengths, limitations, and applicability of each approach. The remaining subsections, including Deep Learning Frameworks, Unified TSAD Models, LLM-based TSAD, and Classical Methods, will be more thoroughly contrasted and summarized in Sec. 7.2.

(2) **Deep Learning Architecture** (Sec. 5.2): This part is one of the main contributions of this work, introducing new mainstream algorithms from the perspective of the model's backbone.

(3) **Unified Time Series Anomaly Detection Model** (Sec. 5.3): Past algorithms (in Secs. 5.1 and 5.2) were only applicable to a single domain and could not generalize to different domains using a unified model with shared parameters. This part introduces the independent multi-domain TSAD model that utilizes domain-shared parameters.

(4) **LLM-based Time Series Anomaly Detection Model** (Sec. 5.4): Applying LLM to TSAD is a new mainstream research approach. The current main routes include fine-tuning based on LLM and using other strategies (such as knowledge distillation, prompting) to stimulate the anomaly detection ability of LLM.

(5) **Classical Machine Learning** (Appendix B.1): Considering the length of the article and the extensive coverage of machine learning applications in TSAD in previous reviews, this section serves as supplementary reading for readers who may be unfamiliar with machine learning.

5.1 Training Method

Following is the presentation of four key training methods for deep learning algorithms in time series anomaly detection. The rationale behind the decomposition of these training methods lies in their core assumptions for identifying anomalies:

(1) **Autoencoder-based Model** (Sec. 5.1.1): This model is trained through dimensionality reduction and reconstruction. Its core idea is to preserve the features of normal samples while distorting abnormal ones.
(2) **Contrastive Learning Model** (Sec. 5.1.2): Leveraging well-designed contrastive strategies, this model establishes clearer decision boundaries between normal and abnormal patterns, thereby facilitating the identification of anomalies.
(3) **One Class-based Model** (Sec. 5.1.3): It assumes that the high-dimensional features of normal time series data are distributed within a sphere, whereas anomalies are located outside of it.
(4) **Generative Adversarial Model** (Sec. 5.1.4): The model can generate both normal and abnormal time series samples simultaneously, which enables it to better recognize anomalies.

5.1.1 Autoencoder-based Model. Autoencoders have become a prominent method for detecting anomalies in time series data without labels by learning compressed representations of normal patterns, facilitating dimensionality reduction and feature extraction. Early explorations include **EncDec-AD** [98], which employs a simple LSTM encoder-decoder scheme. Building on this, **MSCRED** [171] enhances anomaly detection by utilizing system feature matrices for global state representation and a convolutional encoder for sensor correlation extraction, along with an attention-based ConvLSTM to capture cross-temporal information across channels.

Further advancements are seen in the univariate fully connected network (**UAE**) proposed by Garg et al. [37], which introduces dynamic scoring functions that outperform static ones in multivariate settings, particularly when modeling channels separately.

The unsupervised **G-LSTM-AE** model, developed by Hu and Xia [54], combines Gaussian confidence intervals with LSTM for rapid detection in industrial signals, meeting real-time requirements. Additionally, **USAD** [8] features a dual-decoder network structure, employing two-stage training to optimize performance.

Novel approaches, such as **RE-ADTS** [7], use autoregressive models for optimal window width determination, while the **NPSR** algorithm [74] introduces a unique modeling technique that combines point-based and sequence-based reconstruction to quantify anomalies, incorporating a Nominality Score to derive an Induced Anomaly Score. Collectively, these methods illustrate the diverse strategies employed in leveraging autoencoders for effective anomaly detection in time series data.

Additionally, usually the variational autoencoder can be regarded as a special case based on the autoencoder. Therefore, this part is introduced in Appendix B.2.1.

5.1.2 Contrastive Learning Model. Contrastive learning is a form of self-supervised learning that compares different views of data, such as normal and anomalous states, to differentiate their features. In the context of TSAD, this method contrasts the representations of normal time series data with those of anomalies, training the model to recognize patterns of abnormal behavior. This approach effectively captures the underlying structure and variations in the data, enhancing the model's ability to identify anomalies.

In the realm of TSAD, recent advancements leverage contrastive learning to enhance model performance. Notably, **PatchAD** [186] employs a multi-scale patch-based MLP-Mixer architecture, integrating dual project constraints to prevent model collapse. Similarly, **CARLA** [27] utilizes a two-stage process, injecting anomalies during pre-training to create diverse samples and distinguishing normal from anomalous windows.

Continuing this trend, **SimAD** [185] adopts a dissimilarity-based framework with an advanced feature extractor and the ContrastFusion module, enhancing anomaly detection while introducing new evaluation metrics.

On another front, **TimeAutoAD** [69] presents an autonomous framework that utilizes self-supervised contrastive loss and automates hyperparameter optimization, demonstrating robustness against contaminated data. Meanwhile, **USD** [90] captures complex patterns using multi-Gaussian representations, enhancing representation space through data augmentation and soft contrastive learning. BroadCAE [183] replaces the reconstruction loss of board learning system with a contrastive loss. It learns portions of normal and anomalous samples in a low-dimensional random space, while simultaneously training a dynamic threshold selection model capable of generating dynamic thresholds based on input samples.

5.1.3 One Class-based Model. One-class classification methods operate on the premise that normal data points cluster densely, while anomalies are situated further away, allowing for boundary establishment based on normal data distribution. Various innovative approaches have emerged to enhance anomaly detection.

Mauceri et al. [99] introduced a dissimilarity-based representation for time series, transforming them into vectors of dissimilarities from prototypes and utilizing a one-class nearest neighbor for classification. In contrast, **OC-TCN** [103] employed binary cross-entropy as a loss function and integrated transfer learning to facilitate cross-domain knowledge transfer in TSAD.

THOC [120] utilized an RNN with skip connections to capture multiscale features and developed a two-stage feature fusion scheme for TSAD. **COCA** [143] focused on data augmentation techniques, including jittering and scaling, while implementing a contrastive learning strategy that operates without negative samples, addressing model collapse. **COUTA** [156] emphasized calibrating irregular noise and enhancing the significance of normal samples through uncertainty modeling, introducing a one-class loss based on prior distribution to improve prediction reliability.

5.1.4 Generative Adversarial Model. Generative Adversarial Network (GAN) have emerged as a powerful tool in TSAD by training a generator to replicate normal data patterns while a discriminator learns to differentiate between generated data and actual anomalies. This adversarial mechanism enhances the understanding of normal patterns, thereby improving anomaly detection in dynamic time series. Several approaches [33, 93, 105, 169] have been developed within this framework.

FGANomaly [33] utilizes both reconstructed and original time series data as inputs to its Filter module and discriminator, incorporating a progressive Adaptive Weighted Label function to enhance optimization during training.

In a different approach, Niu et al. [105] combined LSTM networks with VAE to learn the deep probability distribution of time series, applying GAN principles to assess whether real-time data fits the normal distribution using both reconstruction and discriminator errors.

The integration of advanced architectures is also evident. Zeng et al. [169] introduced a GAN with Transformer architecture, leveraging the discriminator to capture short-term trends, which amplifies anomaly scores. **TadGAN** [39] employs LSTM at its core, combining reconstruction loss with three auxiliary functions to refine anomaly discrimination, while Cycle Consistency Loss is included to prevent model collapse.

AnoFormer [124] enhances normal and anomalous data modeling through a two-step masking method, adjusting decision boundaries by calculating uncertainty using entropy. **BeatGAN** [93] merges CNN and RNN structures, optimizing parameters by minimizing reconstruction and discriminator errors, and employs Extreme Value Theory to generate thresholds for anomaly detection without making assumptions about data distribution.

UAC-AD [89] presents a GAN-based method for multimodal data, utilizing the mixup technique to create mismatched samples, thereby improving difficult sample recognition and anomaly detection accuracy through contrastive learning, while optimizing the discriminator via adversarial learning.

Next, we conduct a comparative analysis of the different training methods, summarizing their respective strengths, limitations, and applicability:

Strengths. (i) Autoencoder-based (AE-based) training is the simplest approach, with its dimensionality reduction capability effectively mitigating redundant features. **(ii)** Contrastive learning offers a highly flexible, plug-and-play training paradigm that guides models to learn semantically meaningful features by imposing diverse constraints on intermediate representations. **(iii)** One-class-based training explicitly models the anomaly detection problem by assuming normal data consistently resides within a hypersphere, thereby simplifying the training process. **(iv)** Generative models, including GAN and diffusion models, provide a special capability: they can not only discriminate anomalies but also synthesize both normal and abnormal data.

Limitations. (i) A significant limitation of AE-based models is that their fundamental reconstruction loss often leads to an overemphasis on local features, inadvertently neglecting global temporal patterns. Researchers have addressed this by designing multi-level, integrated, and multi-scale feature extraction methods, as seen in works like [84, 123, 147, 171]. Additionally, [74] introduced an auxiliary sequence reconstruction loss to explicitly encourage the learning of global features. **(ii)** For contrastive learning, a major challenge lies in the meticulous design of proxy tasks to prevent training collapse. Approaches by [27, 69, 179, 185] tackle this by constructing positive and negative samples at the instance level. Alternatively, [157, 164, 186] focus on designing specialized contrastive model architectures as an alternative to explicit proxy task design. **(iii)** One-class-based training suffers from overly strong assumptions, often leading to overfitting where the hypersphere volume shrinks excessively, thereby reducing the discriminative spatial separation of features. To counteract this, [99] introduced prototype features, regularizing the model by requiring features to align with these prototypes. [120] addressed this limitation by incorporating multi-scale feature extraction to expand the feature distribution space. Furthermore, [143, 156] inject calibrating anomalies into time series data during training to increase learning difficulty and constrain hypersphere collapse. **(iv)** GANs inherently suffer from training instability, commonly attributed to the discriminator's superior capacity over the generator. Works like [33, 39, 124] mitigate this by designing novel loss functions to optimize the training process. An inherent drawback of GANs is the need to construct two interacting models, which complicates convergence and increases memory footprint. Although subsequent diffusion models only require a single generative model, they still face challenges in achieving stable convergence.

Applicability. Among all training methods, generative models (including both GANs and diffusion models) generally exhibit poorer practicality. Their higher convergence difficulty and increased model footprint limit their widespread deployment. Furthermore, diffusion models, in particular, often struggle to meet real-time processing requirements due to their inherent computational intensity.

5.2 Deep Learning Architecture

This part classifies the algorithms based on the current mainstream backbones of deep learning. To highlight the contributions of our paper and avoid repetition with previous surveys, the following types of work have been extensively reviewed in other surveys; thus, we provide a detailed discussion in the Appendix B.3:

- **Recurrent Neural Network Models** (Appendix B.3.1)
- **Graph Neural Network Models** (Appendix B.3.2)
- **Temporal Convolution Network Models** (Appendix B.3.3)

5.2.1 State Space Model. State Space Models represent a significant class of state-based sequential models, with notable implementations such as Mamba [80]. This survey primarily focuses on deep SSM methods for TSAD, specifically Mamba (or selected state space model), which can be broadly categorized into two types: (i) **forecasting Mamba** [17, 86, 146, 159], and (ii) **TSAD-oriented Mamba** [49, 51]. Mamba is characterized as an S4 model that employs a selection mechanism computed through a scan operation. By refining the state selection equation, Mamba achieves linear complexity, thus facilitating the learning of complex long-range dependencies while minimizing computational load.

Forecasting Mamba: Building on the Mamba architecture, **S-Mamba** [146] is a simplified state space model based on Mamba, designed for time series prediction tasks. Empirical comparisons with state of the art Transformer models show that S-Mamba requires less training time and GPU memory, making it especially suitable for TSAD tasks with strict real time requirements. Several works [17, 86] have built upon the Mamba architecture to propose models better suited for long time series modeling. For example, **Mambaformer** [159] integrates the SSM with the Transformer architecture to create a hybrid Mamba-Transformer framework, showcasing scalability for large-scale long-short range time series forecasting tasks.

TSAD-oriented Mamba: DMamba[18] combines Mamba with a multi-stage detrending mechanism. This integration utilizes SSM's capacity to capture long-range dependencies while the detrending process stabilizes inputs for the sequence model, thereby enhancing generalization capabilities for non-stationary data.

SGFM [49] comprises a feature extraction module and a conditional flow matching module for TSAD. The feature extraction module leverages state space models to capture long-term temporal dependencies and graph convolutional networks to model spatial relationships within the time series data. The extracted features are then used as conditions to guide the flow matching process, which models the normal data distribution by transforming a simple prior distribution to the target distribution through learned vector fields.

TSAD-C [51] introduces a Decontaminator that combines masking strategies with an SSM-based conditional diffusion model. By leveraging an SSM to capture intra-variable dynamics and a GNN to model inter-variable relationships, it excels at detecting anomalies that span long temporal horizons.

ASSM [170] is a streaming TSAD model that suppresses redundant noise by dynamically adjusting hidden features, thereby accelerating the response to abrupt anomalies. It is tailored for resource-constrained sensor environments.

Lastly, **MemMambaAD** [80] first decomposes the time series into trend and seasonal components, then employs the proposed Mamba-TCN module to extract features from each component separately. A dynamic memory bank is introduced to continuously store the decomposed representations of normal samples, enhancing TSAD performance.

5.2.2 Diffusion-based Model. Diffusion models simulate a physical process in which data points transition from an ordered state (normal data) to a disordered state (anomalous data). In the context of TSAD, these models are trained to replicate the generation process of normal data. After training, anomalies can be detected by analyzing the model's behavior during the generation phase. This approach offers significant flexibility and controllability over the data generation process, making it particularly effective for managing complex time series datasets.

A notable contribution in this area is **D3R** [140], which tackles the instability of time series data in non-stationary environments, a challenge that often results in a high false positive rate. D3R employs a decomposition and reconstruction method to manage data drift, integrating a noise diffusion model that directly recovers contaminated data. This strategy allows for avoiding the substantial retraining costs typically associated with changes in information bottlenecks. Building on this foundation, **ImDiffusion** [19] presents a TSAD model that merges time series imputation techniques with diffusion models. It utilizes a grating data masking strategy to generate missing data points, enhancing the decision boundary between normal and anomalous data. Research indicates that this imputation approach is more effective than traditional prediction and reconstruction methods in improving model performance.

Additionally, Pintilie et al. [112] propose a novel use of diffusion models specifically for TSAD, introducing DiffusionAE, which is a jointly trained autoencoder and diffusion model. This model conditions the diffusion process on the autoencoder's reconstruction, demonstrating that training with reconstructed data results in a more robust model compared to using the original data.

Another significant advancement is **DDMT** [162], which introduces the Adaptive Dynamic Neighbor Mask (ADNM) mechanism to address the Weak Identity Mapping issue in time series reconstruction. ADNM dynamically adjusts the mask based on the differences between the reconstructed and original time series, thereby enhancing the model's ability to understand and fit normal data patterns.

Finally, **TimeADDM** [55] combines an RNN embedding module with the denoising capabilities of diffusion models for TSAD. The process begins by mapping the original time series to a low-dimensional latent space, followed by introducing noise through a forward diffusion process and reconstructing features via reverse denoising. Anomaly scores are computed based on reconstruction errors across various diffusion steps. Similarly, **DDTAD** [131] utilizes a TCN for pre-embedding time series data, incorporates a one-dimensional UNet diffusion network for data restoration, and subsequently detects anomalies.

5.2.3 Transformer-based Model. Transformer-based models have revolutionized the field of time series anomaly detection by leveraging a self-attention mechanism, which enables the processing of sequential data in parallel while being independent of fixed positional relationships among elements. This architecture excels in capturing long-range dependencies, thereby enhancing the understanding of complex temporal patterns and improving anomaly detection accuracy.

A variety of innovative approaches have emerged, building upon the foundational capabilities of Transformers. **Anomaly Transformer** [157] introduced an anomaly attention module accompanied by a correlation discrepancy mechanism, integrating prior and sequence correlations to refine normal time series modeling. Their findings suggest that while normal time series exhibit greater correlation with global sequences, abnormal instances tend to correlate more with local sequences.

Expanding the capabilities of traditional models, Zhang et al. [173] proposed the **TransAnomaly** model, which incorporates variational learning concepts into a Transformer framework by enhancing the LSTM-VAE and OmniAnomaly models. They also introduced a gated transition function, significantly improving the model's ability to capture dependencies among hidden variables.

To make anomalies more interpretable, CauseFormer [182] introduces a rule-based tree-structured attention mechanism that simultaneously harvests high-order features and quantifies each metric's contribution, delivering both sample-level and metric-level explanations.

In a bid to reduce computational complexity, Doshi et al. [32] developed **TiSAT**, which features a probability attention mechanism tailored for anomaly detection. Meanwhile, TranAD [137] introduced a purely adversarial generative network based on Transformer architecture, where two decoders compete to enhance anomalies for detection purposes.

AnomalyBERT [64] further advanced the field by utilizing 1D relative positional bias embeddings along with cross-entropy for prediction loss. They implemented four data augmentation techniques—Soft replacement, Uniform replacement, Length adjustment, and Peak noise—demonstrating how self-supervised classification can enhance model adaptability across various datasets.

Moreover, **DCdetector** [164] proposed a patch-based dual self-attention representation learning mechanism that effectively captures normal sequence feature representations through both global and local patches. Their multi-scale representation mechanism mitigates information loss during downsampling.

Lastly, the **D3TN** model [139] showcases innovative feature and temporal modules, utilizing graph convolutional networks and self-attention for extracting both global and local features. Additionally, it incorporates the POT-MoM algorithm for determining the final anomaly threshold. In a different approach, Kim et al. [72] introduced a Transformer-CNN hybrid anomaly detection network, employing attention mechanisms for multi-layered feature extraction complemented by CNNs for anomaly prediction.

5.2.4 Prototype-based Model. The prototype (or memory bank) mechanism serves as an effective technique for storing features, thereby reducing the number of training parameters by retaining a subset of them. This method mitigates the limitations faced by temporal models, which often struggle to accurately memorize the representations of all normal data. By integrating prototypes into the model, the capability for feature learning is significantly enhanced, facilitating the differentiation between normal and anomalous data.

Several studies have successfully implemented this prototype mechanism. For instance, **MEMTO** [128] utilizes a memory-guided Transformer that markedly boosts performance through the incorporation of a gated memory module, a two-phase training paradigm, and a novel bidirectional deviation anomaly score grounded in Latent Space Deviation. This approach underscores the effectiveness of memory integration in improving model robustness.

Similarly, **AMSL** [179] addresses the challenges posed by insufficient normal data and feature representation by integrating six time series data augmentation methods. It leverages self-supervised learning in conjunction with memory networks, proposing a global-to-local memory network architecture that enhances the final feature representation. This illustrates the potential of memory networks in augmenting data representation.

Moreover, **AdaMemBLS** [184] combines the rapid inference capabilities of broad learning systems (BLS) with memory prototypes to effectively distinguish between normal and anomalous data. This approach not only offers faster processing speeds compared to deep learning methods but also maintains competitive performance. AdaMemBLS enhances the model's capability to learn time series features through various data augmentation techniques and incremental learning algorithms. Additionally, it employs diverse ensemble methods and discriminative anomaly scores to bolster anomaly detection performance. The theoretical analysis presented in the paper further elucidates the algorithm's time complexity and convergence, reinforcing the viability of the prototype mechanism within BLS.

5.2.5 Others. In TSAD, some algorithms utilize flow-based neural network architectures, like GANF, to improve data distribution estimation. Ensemble-based algorithms aggregate multiple base models in various ways to address TSAD challenges with a more extensive model set. Furthermore, researchers propose integrating federated learning with TSAD to meet high privacy and security demands.

Dai and Chen [24] presents a novel framework called Graph-Augmented Normalizing Flows (**GANF**) for TSAD. This framework identifies distribution drift by incorporating Bayesian networks to model conditional dependencies among variables while jointly estimating flow parameters.

A prior study [62] significantly advances anomaly detection by introducing an ensemble of deep learning models for multivariate time series data. By combining RNN, LSTM, CNN, Transformers, and GNN models, the detection of anomalies is greatly improved. This ensemble approach harnesses the strengths of various algorithms to enhance accuracy and resilience, especially in real-time processing and decision-making in diverse industrial contexts.

Xu et al. [158] introduces **PeFAD**, a Parameter-Efficient Federated Anomaly Detection framework tailored for time series data. Aimed at providing privacy-preserving and scalable anomaly detection in decentralized environments, PeFAD employs a LLM to tackle data scarcity issues and enhance the model's ability to generalize from limited data.

Moreover, [65–68] fuse fuzzy theory with ensemble learning to devise lightweight, online TSAD algorithms tailored for IoT systems.

5.3 Unified Time Series Anomaly Detection Model

The Unified Time Series Anomaly Detection Models adopt a unique strategy, enabling a single model with shared parameters to manage multiple distinct domains of time series data independently. Several recent models have advanced the field of TSAD through innovative mechanisms and cross-domain adaptability. **DADA** [121] represents a universal TSAD model capable of zero-shot anomaly detection without requiring fine-tuning on specific datasets. This capability is achieved through adaptive bottleneck and dual adversarial decoder mechanisms, which enhance its generalization across diverse downstream scenarios. DADA's adaptive bottleneck module creates pools of varying latent space sizes tailored to the information bottleneck requirements of different datasets, while its dual adversarial decoder module facilitates the learning of normal and abnormal time series patterns through adversarial training, thereby establishing clear decision boundaries. Complementing DADA's approach, **DACAD** [28] introduces a cross-domain TSAD model that integrates domain adaptation with contrastive learning. By leveraging labeled data from one domain, DACAD enhances unsupervised TSAD tasks. It generates triplets of target data, source data, and labels by injecting anomalies into both domains, utilizing TCN for representation. The model employs supervised average margin contrast loss for the source domain and self-supervised contrast triplet loss for the target domain, effectively distinguishing between normal and anomalous data through adversarial training, and utilizes an extended Deep SVDD structure as the anomaly detector.

In the realm of unsupervised TSAD, **ContexTDA** [75] introduces a context-aware domain adaptation model. It conceptualizes the time series context sampling problem as a Markov decision process and employs deep reinforcement learning algorithms with customized reward functions to facilitate knowledge transfer, thereby enhancing anomaly detection capabilities. **FS-ADAPT** [82] specifically targets few-shot learning and unsupervised domain adaptation. Its dueling triplet network processes rearranged and augmented quadruplet data while engaging in adversarial training of feature extractors. FS-ADAPT effectively extracts class-specific and domain-specific features to learn domain-invariant representations, addressing concept drift in anomalies through incremental adaptation. By maintaining exemplar sets and dynamically updating the feature extractor, it enhances online detection performance amidst challenges posed by limited labeled data in source domains and unlabeled data in target domains.

VANDA [48] exemplifies a domain-adaptive unsupervised model for TSAD that utilizes trainable embedding vectors to represent features of each time series variable. By constructing a graph structure based on these embeddings and engaging in self-supervised graph training, VANDA enhances embedding learning. It leverages attention mechanisms to aggregate neighboring representations and employs techniques like sliding similarity and optimal transport distance to facilitate the transfer of inter-domain variate associations.

UNITS [36] further advances cross-domain time series analysis through a modified transformer architecture. By utilizing three token types—Sample tokens for embedding heterogeneous multivariate time series, Prompt tokens for task-specific context, and Task tokens for TSAD. UNITS efficiently detects anomalies via reconstruction errors. The model's two-way time and variable self-attention mechanism, along with a dynamic linear operator in the feedforward layer, optimizes computation and captures complex relationships in time series data.

5.4 LLM-based Time Series Anomaly Detection Model

Predictive LLM-based Models concentrate on fine-tuning or aligning LLMs into unified time series models for effective TSAD. Conversely, Exploring LLM-Powered Models employ techniques such as knowledge distillation, knowledge injection, and prompt guidance to adapt LLMs for TSAD tasks, leveraging their strong text generation capabilities while addressing their shortcomings in task-guidance mechanisms.

5.4.1 Predictive LLM-based Model. Recent advancements in TSAD have been significantly influenced by the integration of pre-trained models and large language models (LLMs). **TS-Bert** [25] is a pioneering pre-trained model specifically designed for TSAD. It leverages a substantial volume of unlabeled time series data to learn temporal behavior features, employing pre-training tasks analogous to those used in BERT, such as Masked Language Modeling and Next Sentence Prediction. Following pre-training, TS-Bert integrates the input and output of the TSAD task, allowing for end-to-end fine-tuning of all parameters based on target data. Notably, to alleviate the dependency on labeled data, the model incorporates unsupervised training by generating labels within the training data using self-reconstruction during the fine-tuning stage.

Building on this foundation, **GPT4TS** [188] introduces a large language model tailored for general time series analysis, based on pre-trained architectures like GPT-2. It begins by applying Instance and Norm Patching to the input time series data, followed by linear probing to redesign and retrain the Input Embedding. This projection minimizes the parameter count while adapting the model for downstream tasks, including TSAD. During training, GPT4TS freezes self-attention layers and feedforward networks containing extensive pre-training knowledge, fine-tuning only the positional embeddings and layer normalization, which effectively balances the use of pre-trained knowledge with task-specific adaptations.

Similarly, **aLLM4TS** [13] exemplifies a class of LLMs focused on general time series analysis tasks. This model adapts large language models through a two-stage self-supervised multi-patch prediction task, enhancing time series representation learning. By utilizing a unique patch-level decoding layer, aLLM4TS improves its sequential representation capabilities for time patches, demonstrating robust performance across various downstream tasks, including TSAD.

Timer [94] further advances this space by presenting a class of large language models designed for time series analysis, based on a decoder-only Transformer architecture that generates the next token autoregressively. Timer aggregates and cleans public time series datasets to construct a high-quality, large-scale Unified Time Series Dataset, which encompasses heterogeneous domains and hierarchical capabilities. Pre-training on this diverse data, Timer reformats it into a unified single-series sequence format, promoting channel independence. For TSAD tasks, Timer frames the problem as predicting the next token based on prediction errors, effectively achieving its objectives in anomaly detection.

5.4.2 Exploring LLM-Powered Model. Meanwhile, several innovative TSAD models leverage knowledge distillation and LLMs to enhance detection capabilities. **AnomalyLLM** [87] employs a knowledge distillation approach to identify temporal anomalies by training a student network to replicate the behavior of a teacher network derived from a pre-trained LLM, specifically GPT-2. This model addresses the challenge of limited access to large-scale time series datasets by capitalizing on the representation differences between the teacher and student networks. By fine-tuning the positional embedding layer and layer normalization of the student network, AnomalyLLM improves its temporal representation. To mitigate overfitting to abnormal features, it incorporates prototype signals and utilizes data augmentation techniques to synthesize anomalies, thereby increasing the representation gap between the two networks.

LLMAD [91] introduces an LLM framework aimed at enhancing the interpretability of TSAD. This model utilizes the In-Context Learning mechanism to retrieve the top-k most similar normal and abnormal samples from a pre-constructed database. By incorporating this information into the prompts sent to the language model, LLMAD effectively establishes boundaries between normal and abnormal patterns. Additionally, it enhances both performance and interpretability through Domain Knowledge Injection and Expert Step-Wise Inference via the Anomaly Contextualization technique. LLMAD ultimately generates natural language reports that aid decision-making by providing explanations of detected anomalies, classifying anomaly types, and determining appropriate alarm levels.

Further advancing the field, **SIGLLM** [6] presents a zero-shot large language model framework tailored for TSAD, built on architectures like GPT-3.5 and MISTRAL. This framework preprocesses raw time series data into a compatible format using techniques such as Scaling, Quantization, Rolling windows, and Tokenization. It features two detection branches: PROMPTER, which merges input prompts with time series data to assist the model in identifying anomalous indices, and DETECTOR, which leverages the model's predictive capabilities to detect anomalies based on prediction errors. Preliminary results indicate that DETECTOR generally outperforms PROMPTER in detection accuracy, while PROMPTER offers greater cost efficiency.

Lastly, **ParInfoGPT** [106] is a two-stage TSAD model that assesses the reliability of rotating machinery using partial information. In the first stage, the model performs self-supervised reconstruction on a substantial corpus of unlabeled time series data to extract key features. It employs an information masking strategy based on mutual information to identify critical patches for zeroing, thereby capturing essential temporal information. The data undergoes processing through a GPT with embedding layers and a GPT encoder for feature extraction, followed by reconstruction using a feed-forward neural network (FFN). In the second stage, ParInfoGPT conducts Weakly Supervised Classification utilizing a small labeled dataset. By initializing the GPT encoder with parameters from the first stage, it effectively extracts features and fine-tunes classification to infer fault types based on the learned feature representations.

6 Evaluation Methods

Several measures for TSAD have been proposed to quantify the quality of AD methods. We categorize these measures into threshold-based and threshold-independent measures. In this section, we first introduce these two major categories, and then we will list key features of a fair metric for TSAD.

6.1 Threshold-based Metrics

In time series anomaly detection, models output an anomaly score S_i for each point p_i in the time series T. This score indicates the degree of abnormality, with higher values signifying more abnormal points. To evaluate the performance of these models, threshold-based metrics are employed, which involve setting a threshold to classify points as normal or anomalous.

Thresholding functions. Various methods have been proposed to determine optimal thresholds in anomaly detection, aiming to maximize the F1-score and enhance detection accuracy. $\mathrm{F1}_{\mathrm{best}}$ [155] calculates the threshold yielding the best F1-score by enumerating all possible thresholds. The top-k threshold [190] predicts k points as abnormal, matching the actual number of anomalies in the test set. Hundman et al. [59] introduces unsupervised dynamic error thresholds that identify the value causing the greatest percent decrease in the mean and standard deviation after removing samples above it. The tail probability threshold [3] assesses anomaly likelihood based on the complement of the Q-function applied to the mean and standard deviation of the recent short-term average of prediction errors. In addition, thresholding methods based on Extreme Value Theory (EVT), such as Peaks Over Threshold (POT)[21] and its online variant SPOT [126], offer statistically grounded and dynamically adjustable thresholds by modeling the tail behavior of anomaly scores. In summary, these methods provide diverse approaches to threshold determination in anomaly detection, each utilizing different statistical properties and optimization techniques to improve prediction accuracy and reliability.

6.1.1 Point-Based Metrics. **Precision, Recall, and F1 Score** are key metrics in TSAD. In point-based anomaly detection, a predicted point within an anomaly sequence is considered a True Positive. Precision measures the ratio of true positives to total actual positives, while Recall assesses the ratio of true positives to total identified positives. The F1 score is the harmonic mean of Precision and Recall [167].

The point-based metric views each data point as an independent sample, ignoring the temporal aspects of anomaly segments, which limits its effectiveness for sequence anomaly detection. To address this, range-based metrics treat the entire range of anomalies as a single event. Additionally, statistical analysis methods evaluate model performance using foundational metrics [30, 35, 38, 102, 152].

6.1.2 Range-based Metrics. **Point Adjustment Metrics** (or event-based metrics) introduced by Xu et al. [155] provide an intuitive approach by treating each contiguous anomaly sequence as a single event. This method recognizes that detecting just one anomaly within a range is often enough to alert experts. Thus, every timestamp in an anomaly segment receives the highest anomaly score from that segment. The segment is deemed detected if this score exceeds the threshold. The point-adjusted F-score is then calculated using a selected anomaly threshold.

The point adjustment metric has a fundamental flaw: it assigns high scores to random guess models. This occurs because errors are penalized only once, while the entire anomaly sequence is rewarded for detection [73**?**]. To address this, Garg et al. [38] propose the composite F1 score, which combines point-based precision and event-based recall, promoting the detection of at least one point in every anomaly sequence while discouraging incorrect predictions. Additionally, Si et al. [125] introduce a Reduced-length PA that balances point-wise and event-wise detection, utilizing a severity coefficient $ln(k)$ as the reduced sequence length, where k is the anomaly sequence length. Similarly, the PA%K protocol [73] considers a segment detected only if the ratio of correctly detected points in the sequence exceeds a specified threshold.

Position Relativity between anomaly sequences and detection results is a crucial measurement for TSAD that point adjustment metrics overlook. Xu et al. [155] employs average alert delay, defined as the time difference between the first point and the first detected point in the anomaly segment, to measure position relativity. Range-based Precision and Recall metrics (R-based) [135] effectively calculate $Recall_T$ and $Precision_T$ based on the existence and overlap in anomaly sequences. These metrics penalize fragmented predictions for a single anomaly while rewarding detections that are close to a defined relative position to the anomaly. The Time Series Aware Precision (TaP), Time Series Aware Recall (TaR), and Time Series Aware F1 (TaF1) metrics, collectively known as TaPR [60], and their enhanced version, eTaPR [61], evaluate both the detection and the portion scoring of anomalies. These metrics assess whether any part of an anomaly is detected and reward methods that capture a larger portion of the anomalous sequence. Improved F1 scores can be derived from these variations of recall and precision.

Moreover, Affiliation metrics [58] utilize the duration between ground truth and predictions, providing an intuitive interpretation. These metrics normalize precision and recall against random sampling, quantifying improvements over random predictions. However, Affiliation lacks sensitivity for accurate predictions and does not effectively highlight performance differences among models, demonstrating weaknesses in robustness against random predictions. To address these issues, Zhong et al. [185] propose Normalized Affiliation and Unbiased Affiliation (N&U Affs). The former is a special case of the latter but operates independently of the dataset, enhancing versatility. This metric reweights precision to address some limitations of the original Affiliation metric. Besides, the Proximity-Aware Time series Anomaly Evaluation (PATE) [40] incorporates proximity-based weighting, accounting for buffer zones around anomaly intervals.

6.2 Threshold-independent Metrics

Threshold-based metrics require threshold selection, compelling models to learn domain-specific thresholds from training data. In practical applications, the ratio of anomaly labels often differs from that in the training data, and factors like noise and delays can significantly reduce performance on these thresholds [107, 108]. Additionally, comparing models across different datasets poses challenges. Therefore, statistical metrics that reflect overall model performance across all possible threshold settings, known as threshold-independent metrics, are necessary.

6.2.1 AUC-ROC and AUC-PR. Area Under the Receiver Operating Characteristic Curve (AUC-ROC) and Area Under the Precision-Recall Curve (AUC-PR) are used to assess the performance of an anomaly detection model across all threshold settings. The ROC curve visually represents the diagnostic ability of a binary classifier as the discrimination threshold varies, plotting the true positive rate (TPR) against the false positive rate (FPR) at different thresholds. In contrast, the Precision-Recall curve is particularly beneficial for imbalanced datasets, plotting precision (positive predictive value) against recall (sensitivity or true positive rate) across various threshold values. Furthermore, VUS-ROC and VUS-PR [107] enhance traditional AUC measures by evaluating the volume under the surface formed by the ROC or PR curves as thresholds and buffer lengths change. This approach recognizes the continuous nature of time series data, offering a more robust evaluation, particularly in scenarios involving noise, misalignments, and varying anomaly cardinality ratios.

6.2.2 Statistical Analysis. For comparisons of multiple classifiers across various datasets, statistical analysis [30] is employed in TSAD [38, 108]. The Wilcoxon test [152] is frequently used for pairwise comparisons, allowing for the evaluation of significant differences between two methods across multiple datasets. For a more thorough assessment that considers multiple methods at once, the Friedman test [35] is utilized, followed by the post-hoc Nemenyi test [102], which ranks methods and identifies statistically significant performance differences.

The relationship between threshold-independent metrics and threshold-based metrics lies in their foundational components. Threshold-based metrics redefine which points are detected as anomalies and can be directly converted into statistical metrics. For instance, R-based F1 [135] is calculated as AUC-PR using range-based precision and recall. Thus, threshold-independent metrics can be viewed as an extension or complement to threshold-based metrics, providing a broader evaluation framework.

6.3 Essential Features of Qualified Metrics

Designing effective metrics for evaluating TSAD models requires careful consideration of several key aspects. Unlike static data, time series data presents unique challenges due to its sequential nature, the importance of timely detection, and the presence of noise. Here are the essential features that a qualified metric for TSAD should encompass:

- **Temporal Adjacency Awareness (TAA):** Adapt the metric to the sequential nature of time series data, recognizing the importance of temporal relationships by valuing the proximity between predicted and actual anomalies, treating close predictions as true positives.
- **Early Detection (ED):** Prioritize the early detection of anomalies by assessing how closely detection aligns with the onset of the event.
- **Balance between Existence and Duration (BED):** Ensure the longer and potentially more severe anomalies are emphasized while not dominating the evaluation, thus recognizing both short and long anomalies appropriately.
- **Robust to Noise, Lag, and Imbalance Category Ratio (RNLI):** Develop a metric that is resilient to noise in anomaly scores, can manage delays in detection, and is sensitive to the ratio of normal to abnormal labels, providing a balanced evaluation.
- **Exactly-Once Detection (EOD):** Encourage each anomaly instance to be reported exactly once to avoid duplicate detections, which can lead to repeated alerts for a single event and confusion about whether alerts refer to the same anomaly.
- **Threshold-Free Evaluation (TFE):** Create a metric that does not rely on predefined threshold values for classifying anomalies, ensuring fair evaluation across various methods and datasets.

Inspired by the work of Zamanzadeh Darban et al. [167], widely used and recommended metrics have been integrated for comparison. Meanwhile, equally important metrics required for different scenarios have been appended. Table 3 presents a comparison of the advantages of different metrics and explains their overall characteristics, aiming to provide guidance for metric evaluation. Although metrics for TSAD have been extensively developed, we believe that no single metric is suitable for all scenarios. Therefore, we recommend using multiple different metrics to evaluate models simultaneously. We discuss some issues regarding these metrics in Sec. 7.1.2.

7 Discussions

This part gives an in detail examination of assessment metrics, model selection, and general approaches in TSAD. A detailed view of the current state creates the framework for establishing future directions.

7.1 Discussions on Evaluations

This section discusses the evaluation of TSAD algorithms by examining two critical components: the anomaly threshold and the evaluation metrics. We then present a detailed analysis of various metrics based on our experimental results.

7.1.1 Anomaly Threshold. Thresholds in anomaly detection distinguish normal from abnormal samples. However, many current methods use static thresholds as an invariant criterion to determine whether a sample is anomalous. While this approach is simple and practical, it often becomes inapplicable when facing distribution shifts over time. For example, in financial transaction monitoring, fixed thresholds fail to capture changes in normal behavior patterns caused by market volatility. Although normalization is widely used to align the input distribution $P(X)$, it does not guarantee stability in the model's behavior. Since anomaly scores are determined by the model's learned mapping

Table 3. Comparison of metric features and interpretability.

Metric	TAA	ED	BED	RNLI	EOD	TFE	Overview
Precision							There are strict requirements for low false alarms to ensure the detection of true anomalies as much as possible.
Recall							It pays more attention to detecting all anomalies and tolerates some false alarms.
F1-Score							A delicate balance between precision and accuracy to ensure the overall performance.
F1-PA [155]	✓						It is suitable for situations where there are a few errors in label marking, that is, a small number of biased detection points are acceptable.
Reduced F1 [125]	✓		✓				Similar to F1-PA, but it reduces the influence of biased detection points and evaluate sequence-wise and point-wise anomalies.
PA%K [73]	✓		✓				It is suitable for evaluating sequence anomalies rather than individual points.
R-based F1 [135]	✓	✓	✓		✓		It penalizes the fragmented predictions of the model for interval anomalies, and the continuous prediction results are more human-readable.
TaPR [60]	✓		✓				It is suitable for scenarios where sequence anomalies need to be detected as entire segments as much as possible.
eTaPR [61]	✓	✓	✓				It is suitable for scenarios where sequence anomalies need to be detected as entire segments as much as possible and focuses on detecting longer and more significant sequence anomalies.
Affiliation [58]	✓	✓	✓		✓		It assesses the overall performance of different models and is suitable for comparing and quantifying model performance, but lacks a certain degree of intuitiveness.
N&U Affs [185]	✓	✓	✓		✓		Similar to Affiliation, it reduces the bias of the evaluation metric itself when the proportion of anomalies is high.
PATE [40]	✓	✓	✓	✓		✓	It is suitable for providing evaluation requirements for pre-warning and post-reminding.
AUC-ROC				✓		✓	It evaluates the model's discrimination between normal and abnormal instances (ignoring the sequentiality of the time series).
AUC-PR						✓	It is suitable for unbalanced datasets when anomalies are significantly less than normal data (ignoring the sequentiality of the time series).
VUS-ROC [107]	✓			✓		✓	It is suitable for situations where a extensive evaluation of the model is required without the need for a threshold and the discrimination between normal and abnormal of the model.
VUS-PR [107]	✓					✓	It is suitable for situations where a extensive evaluation of the model is required without the need for a threshold and anomalies are scarce in the dataset.

from input X to output Y, i.e., $P(Y|X)$ [95], shifts in this conditional distribution—caused by temporal variations or contextual changes—can alter the score distribution, making fixed thresholds unreliable.

This highlights the necessity of dynamic thresholds: in many real-world settings, the underlying joint distribution evolves over time, and anomaly thresholds should adapt accordingly. Unfortunately, the current TSAD literature has paid limited attention to this issue. Future research could explore models that determine thresholds in an end-to-end fashion, without relying on manual tuning or optimal fixed values. Such designs would require the model to recognize and adapt to varying scenarios on its own. Particularly if the community aims to develop general-purpose TSAD models, it will be essential for them to dynamically infer scenario-specific thresholds based on context-aware behavior rather than static assumptions.

7.1.2 Flawed Metrics. In TSAD, selecting evaluation metrics is vital for assessing algorithm performance, yet many commonly used metrics have limitations. Previous studies often focused on F1-PA score and other metrics like AUC-ROC and AUC-PR, but research [32, 58, 107, 185] has highlighted the issue of inflated F1-PA, which is unsuitable for evaluating interval anomalies. Although the community has begun exploring more diverse metrics, no single optimal metric exists. Different algorithms often excel in only one metric, making it difficult to guarantee strong performance across all metrics. The absence of unified evaluation standards and benchmark tests in TSAD complicates comparisons among studies.

We first construct a simple case to demonstrate the limitations of the F1-PA score, followed by a detailed analysis of all metrics in the next subsection.

Case Analysis. We conduct a brief analysis of the most common metrics (precision, recall, F1, F1-PA [155], Aff-F1 [58] and UAff-F1 [185]) currently used. Fig. 5 illustrates a hypothetical scenario for detecting anomalies in time series data. The top row of the figure displays the actual labels for the data series. The terms Random1, Random2, Pred1, and Pred2 denote the anomaly scores generated by two arbitrary algorithms and other methods. Labels ending with “PA” indicate point adjustment, meaning that if the model detects any part of an anomaly segment, the entire segment is considered detected.

Random1 flags every time point as anomalous, whereas Random2 randomly and uniformly selects anomalies. Despite this, both methods achieve high F1-PA scores, which suggests that F1-PA scores are not reliable in this scenario.

The F1 score might be problematic because it is better suited for assessing shorter anomalies, such as the one at the end of the series. However, it can still offer valuable insights and serve as a reference point for non-random predictions.

Among the discussed models, Pred1 has the best performance. Nonetheless, its Aff-F1 score is lower than that of Random1, due to the original Aff-F1 metric’s inability to consider false positives. Fortunately, the improved UAff-F1 metric offers a more accurate way to compare Pred1 and Random1. Moreover, Pred1 also outperforms Pred2, albeit with a small margin in Aff-F1 scores, which are even lower than those of Random1. Here, the UAff-F1 metric is more effective, giving scores of 0.1818 to Pred1 and 0.57 to Pred2, while scoring Random1 and Random2 lower.

In conclusion, the F1 score can be a useful benchmark for algorithms that are not based on randomness. The F1-PA score might lead to misleading improvements.

Currently, it is advisable to avoid F1-PA metric and range-wise metrics considered more reliable [116, 185]. Furthermore, our detailed analysis of metrics (Sec. 7.1.3), coupled with the broad consensus within the community that no single metric can effectively evaluate all models across varying scenarios (e.g., datasets with different anomaly ratios and time series lengths) [92, 108, 153, 185], underscores the critical importance of concurrently assessing models using multiple metrics and diverse datasets. Moreover, providing open training and testing code, as well as sharing models and their weights, is paramount for fostering community progress.

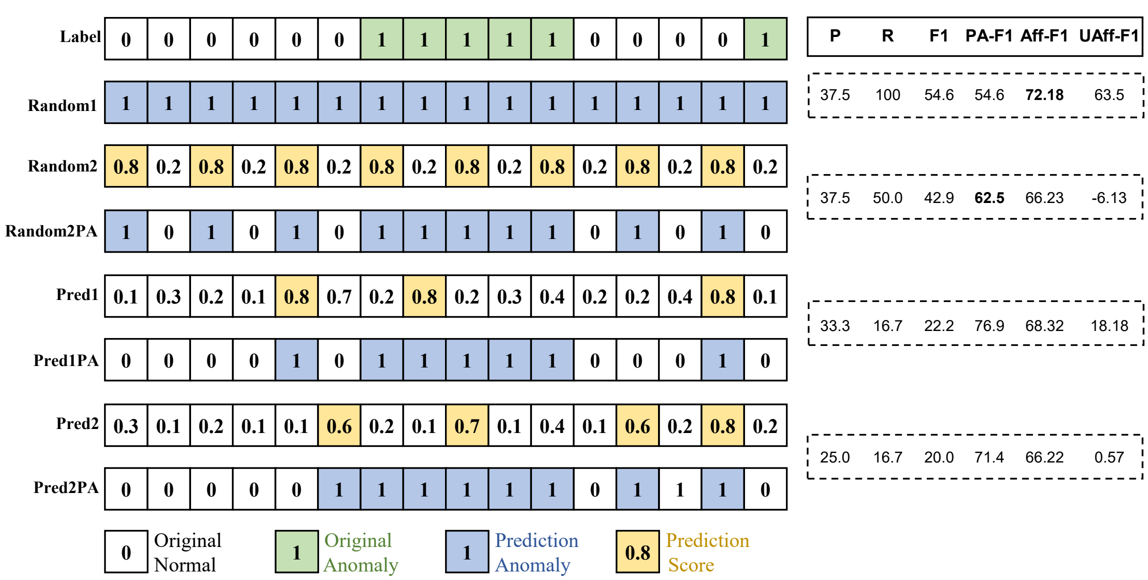


Fig. 5. A case of artificial data illustrating the shortcomings of different metrics.

7.1.3 *Analysis of Metrics.*

In this part, we conduct an experimental analysis of all metrics listed in Table 3. Our focus is on evaluating their performance and analyzing their latency across two scenarios: synthetic data and real-world data.

Dataset Configuration. To evaluate the different metrics, we used both synthetic and real-world datasets. Table S9 details the time series characteristics of these datasets. We generated a total of 30 synthetic datasets, exemplified by names like **100k-20seg-50L** and **100k-20seg-50H**. In these names, "100k" indicates the total length of the time series (i.e., "TS Length" column), "20seg" signifies an expected generation of 20 anomaly segments (i.e., "Segments" column), and "50L" denotes an average anomaly segment length of 50 with low variance (e.g., lengths between 40 and 60). Conversely, "50H" also indicates an average segment length of 50, but with higher variance (e.g., lengths between 1 and 99). The "Max/Min Seg Length" columns represent the maximum and minimum lengths of the generated anomaly segments, respectively. We created 15 synthetic datasets each for time series lengths of 100k and 10k. Additionally, we selected 6 real-world datasets based on their characteristics, such as length, number of anomalies, and anomaly lengths.

Algorithm Configuration. To ensure an objective evaluation of the metrics, we simulated anomaly detection algorithms with varying performance levels instead of using actual algorithms. This approach helps avoid bias that might arise from real algorithms having different adaptability to various data types (e e.g., some algorithms might perform better on smaller datasets or those with a higher anomaly proportion). Specifically, we assumed that an algorithm's prediction accuracy for both normal and anomalous regions is $q\%$, naming such an algorithm Model q, such as Model60 or Model70. For anomalous regions, if the model predicts correctly, the generated anomaly score is $0.1 \cdot U + 0.9$. If it predicts incorrectly, the score is $0.1 \cdot U + 0.05$. Here, U is a random value drawn from a standard uniform distribution. For normal regions, if the model predicts correctly, the anomaly score is $0.1 \cdot U + 0.05$, and if it predicts incorrectly, the score is $0.1 \cdot U + 0.9$. The introduction of a certain level of noise aims to increase algorithm diversity and to measure the robustness of different metrics to noise.

Next, we analyze the variation in model latency and metric performance across several key factors:

(1) the anomaly scoring model used (Fig. 6),
(2) the total number of anomaly segments (Fig. S9),
(3) the average length of anomaly segments (Fig. S10), and
(4) the overall length of the time series (Fig. S8).

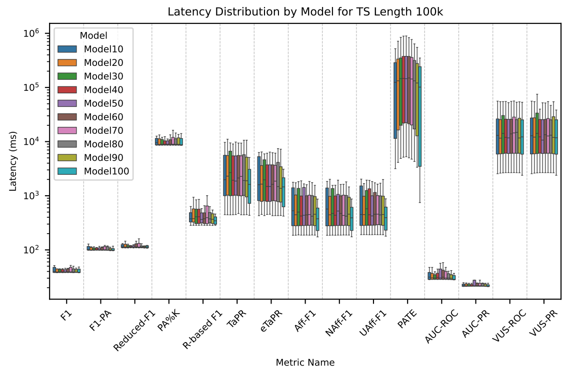
Each factor is evaluated on both synthetic and real-world data. All results are presented visually in figures, with the raw data available in the supplementary materials on our project website.

Performance of Metrics Against Different Anomaly Scoring Models.

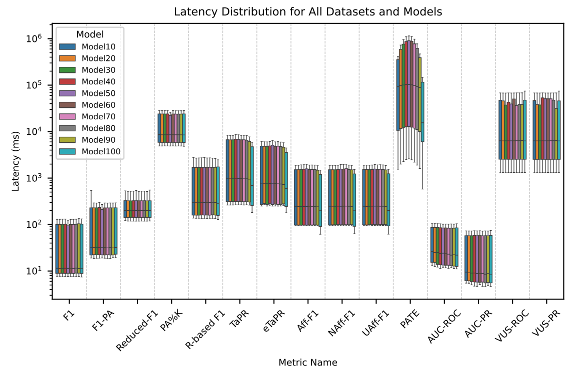
As shown in Fig. 6(a-b), the predictive capability of the model has almost no impact on the latency of most metrics, with only a slight effect on PATE. This is because PATE is influenced by the number of predicted anomaly events (segments). When the model approaches a random model, it generates the most anomaly events, thus resulting in maximum latency.

However, the performance of different metrics varies significantly when facing different models. By analyzing Fig. 6(c-d), we derive the following conclusions:

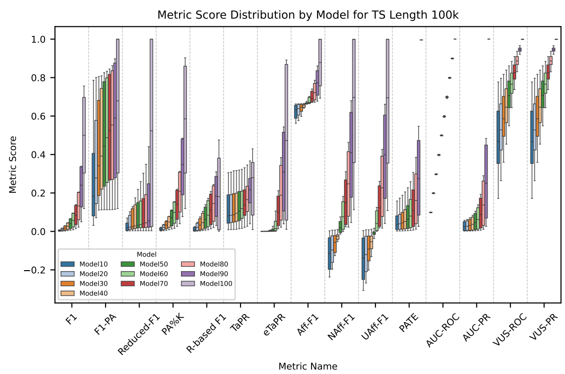
(1) F1 metrics generally show an expected value around 0.5 to 0.6 even when dealing with models with a theoretical accuracy of 100%. This phenomenon occurs because F1 is not robust to noise in anomaly scores. When the anomaly threshold varies, F1 exhibits a large variance, preventing the actual F1 score from reaching its expected value of 1.0.



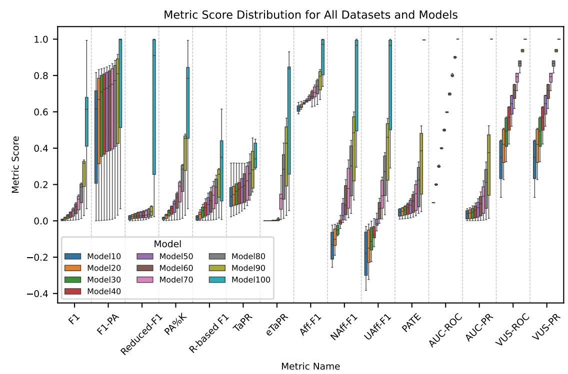
(a) Synthetic data: Latency



(b) Real-world data: Latency



(c) Synthetic data: Performance



(d) Real-world data: Performance

Fig. 6. Performance variation in latency and metrics across different models for synthetic and real-world data.

(2) The F1-PA metric suffers from a significant variance in scores for the same model. This leads to drastic changes in F1-PA when different anomaly thresholds are selected, making model comparison difficult. For instance, in extreme cases, Model100's actual F1-PA score might only be around 0.3, comparable to the level between Model20 and Model30. Conversely, for Model50, the F1-PA score could reach 0.8, exceeding Model100's expected F1-PA score.

(3) Reduced-F1, R-based F1, TaPR, eTaPR, PATE, and AUC-PR are limited by their varying degrees of discriminability across different models. For the nine models with accuracies ranging from 10% to 90%, their metric scores remain low, while Model100's metric score is significantly higher. This issue is most severe for Reduced-F1, PATE, and AUC-PR. For example, PATE's expected score for Model90 is approximately 0.4, but for Model100, it's 1.0. Furthermore, R-based F1 and TaPR fail to reach their theoretical maximum value of 1.0 in both scenarios.

(4) It's important to note that although VUS is a threshold-independent metric, it requires setting an additional event buffer, which can influence the metric's inherent bias. Therefore, when comparing different models, a

consistent buffer size should be used. While buffer size does not affect model ranking, it does impact the absolute metric score.

Impact of Overall Time Series Length.

As shown in Fig. S8(a-b), the overall time series length has a direct and significant impact on the latency of all metrics. The latency for all metrics increases with the growing length of the time series. PATE is the slowest among all metrics; its use is discouraged when the time series length exceeds 100k, as its latency at this point is two or more orders of magnitude higher than that of other interval-based metrics. Among the interval-based anomaly metrics, VUS-ROC and VUS-PR are relatively slower, while R-based F1 is comparatively the fastest.

Fig. S8(c-d) illustrates the sensitivity of all metrics to time series length. Since all anomaly scoring models are expected to have an accuracy of 50%, the theoretical expectation for different metrics should ideally be 0 or 0.5, depending on the metric's inherent characteristics. However, we observed that in synthetic datasets, Aff-F1 exhibited an expected value of 0.67, while VUS-ROC and VUS-PR showed an expected value of 0.8. This suggests a certain bias in these three metrics during evaluation. Although F1-PA's expected value was close to 0.5 in synthetic datasets, its performance on real-world datasets was more concerning. Three real datasets yielded expected F1-PA scores around 0.8, while the remaining three datasets showed expected scores near 0.6, 0.3, and 0. This indicates that F1-PA not only possesses inherent bias but also exhibits a variable bias baseline. Therefore, we do not recommend using this metric in isolation for real-world evaluation scenarios.

Impact of Anomaly Segment Count. Figs. S9(a-b) illustrate how the number of anomaly segments affects metric performance. For synthetic data, we only used time series with a length of 100k (results for 10k or all time series data are in the supplementary materials). With the exception of F1-PA, Reduced-F1, and PA%K, the latency of all interval-based metrics increases as the number of anomaly segments grows. PATE is the most sensitive to the number of anomaly segments; when there are more than 50 anomaly segments in the time series, its latency becomes ten times higher than that of other metrics. If evaluation time is a critical factor, we don't recommend using PATE.

As depicted in Fig S9(c), overall, Reduced-F1, TaPR, PATE, VUS-ROC, and VUS-PR show a clear trend of increasing bias as the number of anomaly segments rises. However, this phenomenon is less pronounced in real-world datasets, largely because these datasets typically contain fewer anomaly segments.

Impact of the Average Segment Length. The average length of anomaly segments has an inverse effect on metric latency compared to the number of anomaly segments. As shown in Fig. S10(a), when the total time series length is fixed, the latency of interval-based metrics decreases as the anomaly segment length increases. Point-based metrics, however, maintain consistent latency regardless of segment length.

The impact of anomaly segment length on different metrics varies significantly. In synthetic datasets (as shown in Fig. S10(c)), the expected value of F1-PA increases with segment length, while the expected values for TaPR, PATE, VUS-ROC, and VUS-PR continuously decrease. Interestingly, when the variance of anomaly segment length is high (e.g., comparing 20L vs. 20H, or 100L vs. 100H), the bias for F1, F1-PA, PA%K, Reduced-F1, TaPR, PATE, and AUC-PR becomes slightly larger. Conversely, NAff-F1, UAff-F1, VUS-ROC, and VUS-PR show slightly less bias under these conditions. This suggests that the former group of metrics is more suitable for short anomaly segments, while the latter group is better suited for long anomaly segments.

Due to space constraints, a more detailed summary and practical recommendations for metric selection are provided in Appendix C.2.

7.2 Discussions on Algorithm

To more precisely delineate the relationships among the taxonomy of paradigms, the models involved in training methodologies, and the architectures of deep learning and machine learning, Fig. 4 illustrates their membership relations. It presents the most prevalent and mainstream instances of these relationships, indicated by arrows of varying colors. The introduction to the remaining algorithms can be found in Appendix B. In this context, these relationships are summarized to facilitate the construction of an overarching panorama of algorithmic innovations. The paradigm is represented as the most general concept, encircled by a red dashed line in the center, while training methods and model architectures are comparatively more specific, delineated by gray dashed lines on either side. The taxonomy of paradigms and their related connections are distinguished by seven distinct colors, while the models of different architectures are differentiated by another five colors ranging from cool to warm. The warmer the color, the stronger the capabilities of the current type of model. In Tables 4 and 5, these different architectural models will be further compared. As shown in Fig. 4, models trained using different training methods are often associated with specific paradigms in the taxonomy. For example, methods based on autoencoders and variational autoencoders are typically grounded in the reconstruction or prediction paradigms. Contrastive learning models and one-class models are often designed based on the similarity paradigm and thus belong to this paradigm. Meanwhile, Fig. 4 also depicts the connections between models and the taxonomy of paradigms from the perspective of the model's backbone or architecture. For instance, current SSMs mostly fall within the reconstruction or prediction paradigms, and their connections with the generation and/or similarity-based paradigms are relatively weak, meaning that there are fewer studies in these areas. Finally, it can be inferred from Fig. 4 that most current TSAD models are of the reconstruction and prediction paradigms, while studies on models of other paradigms are relatively scarce.

To further assist in establishing the understanding of algorithms with different architectures and to comprehend the differences in advantages and disadvantages brought by different architectures, Table 4 provides a detailed summary of the advantages and disadvantages of these algorithms. Meanwhile, based on these advantages and disadvantages, we provide algorithm selection suggestions under different selection criteria, as shown in Table 5. Consistent with the previous Fig. 4 and Table 5, we use different colors to represent algorithms of different architecture types. For example, light gray is used to represent classical machine learning algorithms (ML), and pink is used to represent algorithms with larger capacity such as the unified TSAD algorithm (Unified) and the LLM-based algorithm (LLM). According to different criteria, we categorize these algorithms into three types based on their characteristics. For instance, in terms of the zero-shot ability of the model, it is more recommended to use and study models of the Unified and LLM architectures, but the ML algorithm is not recommended. Meanwhile, other algorithms can be considered as suitable alternatives.

By comparing the strengths and weaknesses of these models, we provide an overall overview and suitable algorithm selection recommendations for researchers and practitioners. Our overarching suggestions are as follows:

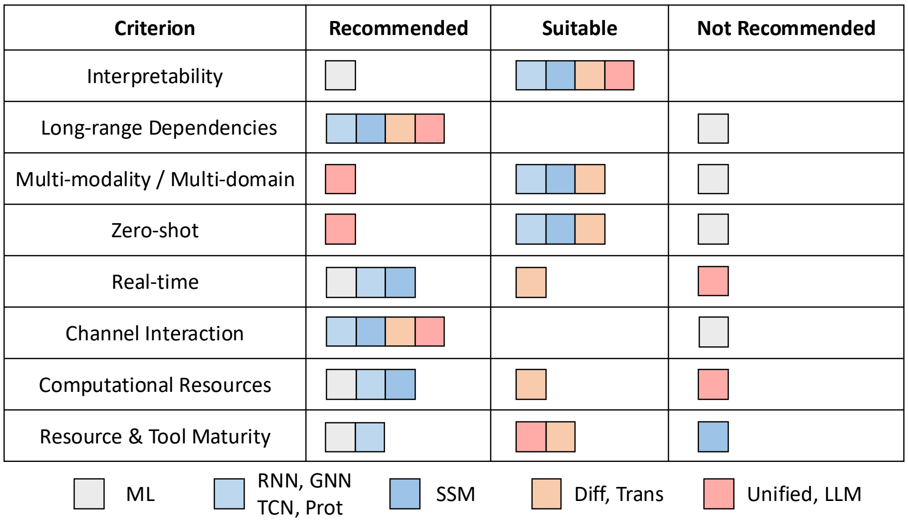
(1) If multi-modality and zero-shot capabilities are essential, prioritize Unified and LLM.
(2) If long-range dependencies and channel interactions are critical, ML models are not advisable.
(3) For interpretability, ML models are recommended.
(4) When factors such as real-time performance, computational resources, and resource & tool maturity are considered, prioritize models such as RNN, GNN, TCN, Prot.
(5) If model performance is the foremost priority, it is advisable to select Trans.

Table 4. The comparison of the advantages and disadvantages of models with different architectures.

Architecture	Advantages	Disadvantages
Unified TSAD Model Exploring LLM-Powered Model Predictive LLM-based Model	①Strong model representational capacity. It serves as an excellent carrier for achieving multi-domain. ②Conveniently combined with other modalities and models to achieve multi-modality. ③Leverage the powerful text generation ability of pre-trained LLM to explore and study more novel detection frameworks.	①Large model size, high space occupancy, and slow computational speed. ②The research is still in its early stage and has not shown a significant gap compared to other models. ③The current mainstream research is combined with the text modality, presenting limitations in usage scenarios.
Diffusion-based Model	①Strong generative capacity. Diffusion models excel at learning complex, high-dimensional data distributions thoroughly. ②High flexibility. Does not have overly strict assumptions on the distribution of data.	①High computational cost. Require multiple denoising steps during both training and inference. Moreover, the inference speed is slow and not suitable for real-time scenarios. ② Large data demand. A large amount of training data is required to achieve good results and capture normal data patterns.
Transformer-based Model	①Transformers are highly effective at capturing dependencies between distant elements in a sequence. ②Transformers can process entire sequences simultaneously, significantly speeding up training.	①The computational cost increases quadratically with the sequence length.
State Space Model	①Linear computational complexity. Can handle longer time series sequences and has a faster computational speed. ②Excellent performance. Under the same model scale conditions, the effect is superior to the Transformer model.	①The model structure is relatively new. Compared to mature model structures such as Transformer, the current research is limited, and reusable resources and tools are scarce. ②The long-range capture capability warrants further investigation. Current research has not fully leveraged the linear complexity advantage of the SSM model.
Recurrent Neural Network Model Graph Neural Network Model Temporal Convolution Network Model	①Strong sequence modeling ability. Naturally suitable for processing sequence data. ②Fast model computational speed. Can design a model with an appropriate scale and powerful capacity under limited resource conditions. ③Combined with the GNN network structure, the model structure advantage is more pronounced in scenarios with strong channel relationships.	①Long-term dependency issue. Although such models can theoretically handle long-term dependency relationships, limited by the model capacity and size, the capture ability for long-range dependency relationships is limited in practical applications. ②Limited generalization ability. Compared to structures like Transformer, the capacity improvement of the model does not directly bring stronger generalization ability. Not suitable for scenarios with complex data relationships.
Prototype-based Model	①It explicitly models normality by representing the normal data as a set of learned prototypes. ②The model architecture is highly flexible, allowing the prototype mechanism to be integrated into various components at different stages.	①The prototype features must be explicitly optimized to enhance diversity and prevent collapse into trivial representations. ② An excessive number of prototypes increases optimization complexity and slows down the model.
Classical Machine Learning Forecasting Model Similarity-based Model Tree-based Model Statistical Method	①Strong interpretability. Tree-based models such as decision trees and random forests, their decision-making processes can be intuitively displayed in the form of a tree, making it easy to understand the decision logic of the model and the importance of features. ②Suitable for scenarios with small amounts of data. In cases where the data feature complexity is not high, machine learning methods have higher robustness and lower data requirements. ③Fast computational speed. Suitable for scenarios with high real-time requirements.	①Limited model complexity. Traditional machine learning methods fail to provide effective modeling and analysis for complex, nonlinear data patterns. ②Traditional machine learning methods demonstrate limited capability in processing feature/channel interactions. They are unable to fully capture and utilize the complex relationships between features in high-dimensional and complex data. ③Dimension curse. As the feature dimension increases, the computational time of the model significantly increases, and the dimension curse problem is prone to occur, leading to a decline in model performance.

(6) Diff, Unified, SSM, and LLM are not suitable as initial scaffolding, as they remain in a developmental phase compared to other types.

Table 5. The algorithm selection suggestions under different selection criteria.



7.3 Discussions on General Technologies

In this section, we discuss two general techniques in TSAD, primarily focusing on their applicable scenarios.

7.3.1 Time Series Data Augmentation. We summarize the commonly used data augmentation methods in TSAD, and past studies [121, 156, 184] have shown that they can effectively enhance the detection capabilities of the model. It is important to note that in this review, we focus on model-agnostic, plug-and-play time series data augmentation methods. We have implemented these methods and will make the code publicly available[2].

Mixup: During training, mix two channels through linear combination to enhance the generalization ability of the training set and reduce the oversmoothing issue in time series prediction caused by distribution drift.[168] Past studies [143, 179, 184] have shown that to improve the effectiveness of anomaly detection, we can also artificially construct some enhanced time series data. **Modify**: Modify the value at random positions in the time series data by introducing noise. **Reverse**: Reverse the time series data from the starting position to the ending position. For a given time series data $\boldsymbol{t} = [t_1, t_2, \cdots, t_N]$, the enhanced data can be obtained as $\boldsymbol{t'} = [t_N, t_{N-1}, \cdots, t_1]$. **Scale**: Multiply the time series data by a certain value to change the scale of the original data. The new time series data can be represented as $\boldsymbol{t'} = \alpha \cdot \boldsymbol{t}$. **Smooth**: Utilize functions such as signal processing to transform the original time series data into a smoother curve. **Noise**: Introducing Gaussian noise into time series data can effectively enhance the model's expressive power by learning not only the incremental features directly. Some work, such as [185], treats the noisy data as negative samples in contrastive learning, making the boundary between normal and abnormal clearer.

Imputation (Masking): Reducing prediction errors implies more accurate modeling of multivariate time series, which in turn improves the performance of anomaly detection. Imputation methods significantly outperform prediction and reconstruction methods, demonstrating the lowest prediction errors. This highlights the superiority of filling methods in self-supervised modeling capabilities, confirming the importance of enhancing self-supervised modeling capabilities in improving the performance of MTS data anomaly detection [19, 162].

[2] https://anonymous.4open.science/w/FTSAD-F1DF/

Moreover, [138] introduced the **Multi-PSCA** method. This technique leverages Dynamic Time Warping to create perturbed time series signals, subsequently using feature decomposition and PCA to reduce dimensionality and obtain the time series' path signature. This method also offers value as a complementary augmented feature for time series analysis.

Anomaly Injection: Researchers [121, 156, 184] have previously investigated the generation of anomalous time series, resulting in advancements in various anomaly injection methods. Recently, GenIAS [26] proposed an anomaly time series generation method based on TCN and VAE, demonstrating that the generated anomalous data significantly enhance model performance. In this paper, we primarily focus on model-agnostic anomaly injection techniques. The following is a summary of several anomaly injection methods that do not necessitate the introduction of additional modules:

- Peak Anomaly: Modifying the value of the last time point to the maximum or minimum value.
- Contextual Anomaly: Similar to point injection, altering the value of the last time point to a local mean; naively, this value can be set to 0.5 (when the data range is 0 to 1).
- Collective Anomaly: Changing the values of a fixed segment of the time series to a constant value. For time series with known distributions, new time series can be constructed based on their periodicity and trends.
- Seasonal Anomaly: Assuming the time series data $\boldsymbol{t} = [t_1, t_2, \cdots, t_N]$ has a fixed period of τ, a segment of time series data (such as $\boldsymbol{t}'_{i:j} = [t_i, t_{i+1}, \cdots, t_j]$) can be rescaled by the period, and then $\boldsymbol{t}'_{i:j}$ can replace the original $\boldsymbol{t}_{i:j}$ to generate new data.
- Trend Anomaly: Similar to the generation method of periodic anomalies, assuming the decomposed trend values of the given time series data are $\boldsymbol{s}^{trend} = [s_1, s_2, \cdots, s_n]$, a segment of trend values can be modified by scaling it with a certain coefficient (α) (e.g., $\boldsymbol{s}^{trend,\prime}_{i:j} = \alpha \cdot \boldsymbol{s}^{trend}_{i:j}$), and then new time series data can be generated using the updated trend.

Although data augmentation methods in time series anomaly detection have been widely developed, several issues need to be addressed and discussed in future research.

For example, the linear mixing nature of Mixup essentially involves interpolation that disrupts temporal causality. The dynamical system characteristics of time series data (such as Granger causality [43]) are distorted by random channel combinations, causing the model to learn spurious correlations.

Additionally, traditional Gaussian noise lacks specificity and is prone to being quickly immunized by the model. Time series noise generation based on adversarial training (such as the adversarial perturbation module of TimeGAN [166]) can construct boundary samples that are more difficult to distinguish, compelling the model to learn more robust feature representations.

In the aspect of anomaly data generation, artificially constructed peak/trend anomalies are overly formulaic, resulting in the model merely memorizing specific anomaly types.

Existing methods enhance univariate variables independently and overlook the dynamic coupling relationships between variables. For instance, the coordinated drift of temperature-pressure sensors can lead to collective anomalies, and independent enhancement disrupts these correlations. Future research on data enhancement could adopt the approach of STG-NF [50], which models the graph attention weights between variables to preserve the overall intrinsic causal connections. Alternatively, it could draw on data generation methods from temporal causal discovery to synthesize data [43].

7.3.2 Channel Independence. Channel independence involves learning representations for each variable (channel) separately during model training on time series data, effectively transforming multi-variable tasks into multiple single-variable tasks. This approach reduces the likelihood of overfitting, as models with channel independence are generally less susceptible to it. In contrast, models with channel mixing often overfit quickly due to increased interactions, which can lead to the learning of noise patterns while potentially overlooking valuable latent interactions [104].

Additionally, models that maintain channel independence can naturally extend to learn cross-channel relationships and utilize multi-task learning loss functions for different channels.

For a unified TSAD model, channel independence is likely crucial, given that these models must manage data from various domains with differing variable counts. However, certain patterns, such as channel-correlation anomalies mentioned in the background section, require insights from inter-variable relationships for effective detection. Consequently, future unified TSAD models must integrate feature fusion techniques (like cross-attention) or adaptive multi-channel processing methods (such as selectively applying single-channel models to multi-channel data).

8 Future and Opportunity

This section addresses the future development of TSAD, focusing on current limitations and necessary future research.

8.1 Interpretability

The current state of TSAD faces significant challenges regarding datasets and evaluation metrics. While advancements are being made, the existing datasets and metrics often lead to misleading progress, and adopting overly complex models is not always the best solution [92, 116, 153]. Furthermore, the TSAD community lacks direct comparative evaluation methods beyond standard metrics, which hinders meaningful development. For practical TSAD scenarios, it is crucial to prioritize model interpretability alongside reliability, ensuring models are as explainable as possible. This approach enhances user understanding of anomaly detection results and decision-making processes, fostering transparency and trust in the system.

Although some past TSAD works have claimed their proposed models are interpretable [46, 134, 182, 189], they generally only indicate a high correlation between specific time series channels and the occurrence of anomalies. They fail to provide more detailed information about the root causes of these anomalies. Moreover, existing surveys [45, 127] have predominantly focused on general-purpose interpretable anomaly-detection algorithms, yet they fall short of addressing the specific needs of time-series data.

Before discussing potential future developments in TSAD interpretability, we first analyze what kind of interpretability is truly beneficial for the TSAD task. Useful interpretability extends beyond simple correlation; it must provide actionable insights for domain experts. A beneficial explanation should answer critical questions such as: *Why is this specific point an anomaly? Which factors or variables led to this anomaly at this time?*

Effective TSAD interpretability requires:

(1) **Temporal and Channel Specificity:** Identifying the exact time windows and multivariate channels that contributed most significantly to the anomaly score.
(2) **Root Cause Analysis:** Providing detailed explanations of the underlying events or behaviors that define the anomaly, rather than just identifying its presence. Specifically, models should analyze the causal processes that led to the anomaly, explaining how changes in certain channels influenced subsequent changes in others, ultimately triggering the anomalous event.

(3) **Contextual Understanding:** Explaining how the anomalous behavior deviates from expected normal patterns and providing context. Models should be able to identify which specific subsequences or shapelets within a channel caused the anomaly. This allows human experts to understand the anomaly's context more clearly, enabling them to efficiently verify (determine if it is a true anomaly) and debug (identify the root cause) the issue.

To overcome the current limitations, the community needs to focus on developing frameworks that support the evaluation of interpretable models:

(1) **Synthesizing Interpretable Datasets:** We can begin with simple, artificially synthesized data to generate datasets based on common time series anomaly patterns. This approach allows for controlled environments where ground truth explanations are known. For example, a synthetic manufacturing dataset could define an anomaly as the result of a specific temporal interaction between multivariate features (e.g., simultaneous spikes in "vibration" and "temperature" channels), requiring models to interpret complex, multivariate relationships.
(2) **Structured Annotation and Labeling:** Structured labeling needs to provide more than just a binary "anomaly" tag. Annotations should categorize anomalies according to their type and explicitly link them to their root cause, providing structured feedback for model training.
(3) **Developing a Structured Language for Root Causes:** A structured language is essential for the rigorous evaluation and comparison of interpretability across different models. For instance, this structured language should be capable of capturing detailed causal processes, including: (i) the type of anomaly, (ii) the temporal position and channel of occurrence, (iii) the root cause channel(s) and corresponding subsequence(s) that triggered the anomaly, and (iv) the indirect channel impacts and subsequence changes resulting from the root cause. This level of detail allows for precise, automated comparison of explanations across models. The purpose of establishing a structured language is to describe anomalies accurately and logically, and to convert these descriptions into a unified or structured database. Such a consistent representation enables standardized evaluation across different models and datasets, facilitating the creation of better benchmarks.
(4) **Advancing Explanatory Evaluation Methods:** We must devise new metrics that measure the quality of explanations, focusing on conciseness and consistency, beyond existing local and global explanation methods [63]. For example, we need to evaluate explanation sparsity (measuring the minimal set of features required for an explanation) and explanation stability (ensuring consistent explanations for similar types of anomalies, even under minor data perturbations) [97, 149].

Thanks to the advancements in LLMs, we can also explore time series anomaly interpretability methods based on LLMs [34]. Although LLMs are still considered black-box functions, their text generation capabilities can aid in the explanation of anomalies, a premise that has been validated in prior work [36, 91, 188]. LLMs can bridge the gap between complex models and human understanding by transforming technical anomaly detection outputs into coherent, natural language explanations, providing a more intuitive view of the detected issues.

8.2 Zero-shot Learning

Zero-shot learning (ZSL) is a paradigm enabling models to identify and detect samples from classes not directly trained on [144]. In TSAD, ZSL allows models to recognize anomaly patterns absent from the training data. With the emergence of LLMs and the trend toward large-scale models, there has been a surge of interest in developing time series large models or specialized TSAD models. Attention has been shifting towards using a unified model to handle diverse time series tasks across various domains [121, 144, 188]. This review highlights several challenges that need to be addressed in ZSL for TSAD:

(1) **Lack of annotated data:** A primary challenge in Zero-shot learning is the limited availability of annotated data for specific classes, complicating the training of TSAD large models. Anomaly detection often suffers from a scarcity of annotated anomalies, with most labels being binary and lacking clear definitions that can vary across datasets or scenarios. This issue hinders the future development of TSAD large models. A potential solution is to inject contextual knowledge into the model through LLMs or other methods to mitigate data scarcity.
(2) **Class inconsistency:** In ZSL, a model is trained with one set of class labels and encounter a completely different set of labels during testing or deployment. This inconsistency requires the model to have high flexibility and adaptability.
(3) **Model complexity:** Achieving ZSL requires more complex model structures to capture generalized features and distinctions among classes, which increases complexity and training difficulty. Current time series models are often limited to fixed-length data or a fixed number of variables, while real scenarios involve varying data lengths and variable counts [36], indicating the importance of developing new time series large models that accommodate the uniqueness of time series data.
(4) **Novel model structures:** Time series models typically require fine-tuning with at least some domain data to adapt to new scenarios [36, 188]. This necessity for a small amount of data, such as 5%, for fine-tuning indicates a significant gap between few-shot and zero-shot learning capabilities.

The development of a unified TSAD model based on zero-shot learning is essential for making large-scale TSAD models commercially viable, similar to how LLMs can perform complex text generation tasks without fine-tuning. TSAD models that lack zero-shot capability are inherently limited in their potential for large-scale applications.

8.3 Multi-Modality Learning

Current TSAD methods based on classical machine learning and deep learning techniques typically focus on a single modality: sequential data. In contrast, unified anomaly detection methods utilizing LLMs often incorporate both sequential and textual modalities directly or indirectly. The first approach involves inputting sequential data transformed into vectors into LLMs. To align these two-dimensional spaces and allow LLMs to comprehend temporal sequence embeddings, existing research typically redesigns the input embedding layer and fine-tunes it on downstream time series datasets. The second approach employs prompt learning strategies to transform sequential data into prompts; however, representing numbers as text reduces the semantic value of numerical data.

Moreover, if time series data can be made interpretable or linked to visual representations, these can be integrated to utilize the intrinsic knowledge embedded in visual modalities or pre-trained visual language models [178]. Insight Miner [101] fine-tunes the visual-textual model LLaVA to create a trend description model that converts windowed time series images into descriptive language. CLIP-LSTM translates stock market data into sequences of textual and visual price representations, using features generated by the CLIP language-vision model for downstream tasks. However, overall, there is limited research on multi-modal TSAD in the community, which presents promising opportunities. Increasingly, studies from various fields highlight the importance of multi-modality for models, reflecting the diverse perceptual modes of humans. Therefore, exploring how to input information from multiple modalities into models and align the embedding spaces of different input modalities is crucial for TSAD.

However, the development of current multimodal TSAD models remains in its early stages [36, 101, 178?]. Existing multimodal TSAD models merely focus on integrating various types of modal data into the overall model learning process, hoping that different modalities can compensate for the inherent shortcomings of time series modalities themselves, such as difficulty in extracting global information and insufficient semantic richness. This approach struggles to foster an agent that truly comprehends the intrinsic relationships within multimodal data. To achieve multimodal knowledge integration, rather than mere feature fusion, we believe the following represent future development opportunities:

(1) **Text Modality for Enhanced Human Understanding:** Current TSAD models lack interpretability. We expect future text modalities to be integrated throughout the multimodal learning process, enabling TSAD models to understand the contextual information corresponding to time series data and articulate the model's understanding. The involvement of natural language can significantly enhance the model's practical usability, finding application in scenarios with higher safety requirements.
(2) **Multi-round Feedback Learning Paradigm:** To further enable natural language to provide feedback to humans, it is necessary to implement a multi-round conversational training paradigm combining time series and text. This would allow the model to ultimately understand time series data by leveraging both text and time series itself, and the model should be able to output both text responses and additional time series data.
(3) **Interactive Visual-Temporal-Textual Programming:** Furthermore, users can real-time constrain model behavior in multi-round conversations using three modalities: natural language, hand-drawn red boxes (visual prompts), and sliding time series windows (temporal prompts). The model needs to support "multimodal prompt compilation": converting hand-drawn boxes into attention masks, and language constraints into differentiable logical rules. This paradigm treats human intent as a fourth modality, enabling interactive anomaly analysis. Under this approach, the model can comprehend the continuous evolution of scenarios.

8.4 Multi-Domain Learning

Unlike conventional domain-specific models that require separate training for data from each domain, a unified TSAD model should share parameters across multiple domains and handle various tasks simultaneously. It should exhibit the following characteristics: (1) Time series data agnosticism, allowing the model to process time series data of different lengths and variables without modifying its structure; (2) Parameter sharing, enabling the model to utilize shared parameters for concurrent task handling [36].

Considering previous discussions, we anticipate that future unified TSAD models need the following characteristics: (1) Zero-shot capability: Current large-scale time series models often require fine-tuning with few-shot data when switching scenarios. Making TSAD models commercially viable will depend on enabling them to perform anomaly detection without any fine-tuning. (2) Dynamic threshold selection: Given the inherent differences in data across domains, the model should adaptively select thresholds based on the current scenario. As discussed, using dynamic thresholds is essential for the current model design. For TSAD models that learn across multiple domains, it becomes even more important—not just to handle fixed thresholds failing over time, but also to deal with the added uncertainty of thresholds varying across domains. (3) Interpretability: Identifying anomalies in time series data is only the initial step. In practical applications, early alerts are essential, but the model must also explain the anomalies to facilitate subsequent tasks, such as identifying root causes and suggesting fixes.

8.5 Foundational Models

The development of foundational models is essential for TSAD, as an effective foundational model is necessary to adapt to diverse TSAD tasks.

Currently, the mainstream paths for time series models can be divided into two categories: unified TSAD models and LLM-based TSAD models.

(1) **Unified (independent) TSAD models:** Research focuses on pre-training foundational models using time series data that can adapt to any domain. As outlined in Sec. 5.3, this approach involves transforming multi-variable time series data into single-variable data, demonstrating some zero-shot capabilities on unknown tasks. However, limited extensive time series data restricts the emergence capabilities of these methods. Additionally, since they rely on a single time series modality, they cannot utilize priors from different domains, necessitating fine-tuning when switching domains.

(2) **LLM-based TSAD models:** Another path for foundational TSAD involves adapting extensively trained LLMs for time series tasks. These methods, discussed in Secs. 5.4.1 and 5.4.2, aim to leverage the knowledge learned from LLMs for time series applications. However, LLMs are not inherently designed for time series or numerical tasks, which results in inadequate pre-training for understanding time series data. Their tokenizers are primarily created for natural language, neglecting the unique characteristics of time series data. The main advantage of these methods is that LLMs possess text generation capabilities, allowing for the analysis of simple causes of time series anomalies.

While these foundational models have achieved success and gained attention, our discussions on multi-modality learning suggest an alternative approach for implementing a unified TSAD model. This approach involves integrating various modal data, including time series, images, and text, to create a versatile perceptual model capable of identifying and understanding complex patterns and anomalies across different modalities.

9 Conclusion

This review addresses various inadequacies in existing surveys from multiple viewpoints, fostering a more unified knowledge by reviewing the historical evolution and present research status of TSAD. By further refining and constructing anomaly taxonomies and paradigm classifications, it provides a more structured approach linked with modern research. We describe and detail numerous approaches, analyzing their respective merits, shortcomings, and practical usefulness, alongside providing advice for algorithm selection. Concurrently, this research provides fresh designs applied in the TSAD area that were previously neglected in past evaluations, including diffusion models, LLMs, and unified models, thereby mitigating flaws of preceding publications. Regarding the evaluation criteria for algorithms, a comprehensive categorization approach is also offered, and varied metrics are thoroughly defined and critically compared. Furthermore, these measures are empirically examined, and assistance is supplied for their sensible selection to assist researchers and practitioners. In the sections on open discussion and future trends, we delve into the current landscape and prospective opportunities within TSAD, encompassing aspects such as metric evaluation, generalizable techniques, model interpretability, zero-shot learning, multi-modal learning, multi-domain learning, and foundational TSAD models. This review analyzes the present limits within current TSAD research and, therefore, provides potential pathways and directions for development, which are vital for the investigation of future TSAD.

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Appendix

This is the appendix of Future of Time Series Anomaly Detection: From Historical Review to Unified Outlook.

A Related Works

To analyze the research trends of different algorithm taxonomies and architectures in current TSAD, we used Google Scholar[3] to search for various types of algorithms. We used a different set of search queries for each category, which can be found on our project website: https://anonymous.4open.science/w/FTSAD-F1DF/. We recorded the number of papers published each year, covering a ten-year period from 2015 to 2025, with data collected up to June 24, 2025. Fig. S7 shows the number of publications for each algorithm type per year, and the raw data is available in Table S6. The left sub-figure is a statistical breakdown based on the taxonomy of different paradigms, while the right sub-figure is a breakdown based on different algorithm architectures. Each class corresponds to a category in Table 4: **Class #1** includes Unified TSAD Model, Exploring LLM-Powered Model, and Predictive LLM-based Model; **Class #2** includes Diffusion-based Model and Transformer-based Model; **Class #3** includes State Space Model; **Class #4** includes RNN, GNN, TCN, and Prototype-based Model; and **Class #5** includes various machine learning methods.

Based on our analysis of the figure, we summarize the following observations:

(1) The publication trends for the **Forecasting** and **Statistical** paradigms have been gradually slowing down since 2022. Similarly, the publication trend for the **Tree-based** paradigm has also slowed since 2023.
(2) The publication trends for the **Generative**, **Similarity-based**, and **Reconstruction** paradigms have remained consistent, indicating that they are still mainstream research paradigms in current TSAD studies.
(3) Publications related to the **LLM-based** paradigm have been increasing since 2024. As of the data cutoff date, the publication count for 2025 is approximately 2,200, which is already 84.6% of the total for 2024. This represents the fastest growth rate in publication count among all paradigms, suggesting that this paradigm is receiving relatively widespread research attention.
(4) The publication trends for different algorithm architectures are similar, which indicates that algorithms across these architectures are equally popular.

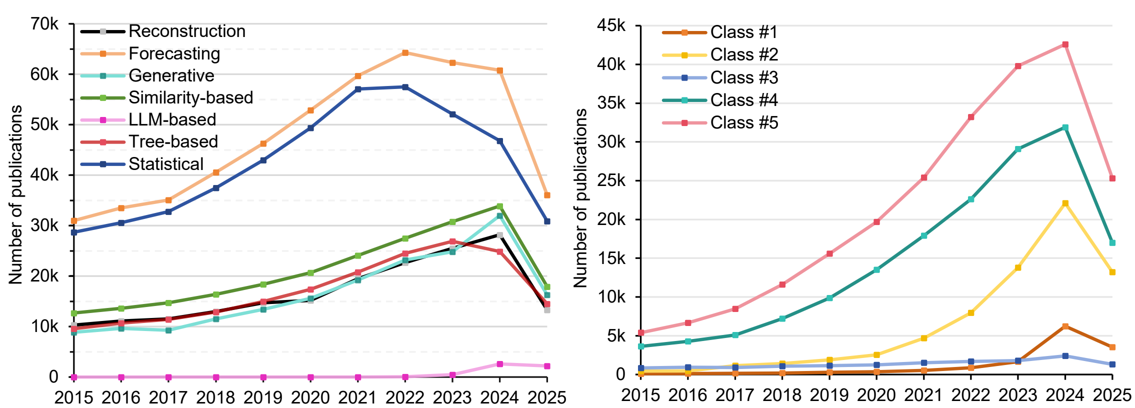


Fig. S7. Publication trends of various algorithm types.

[3] https://scholar.google.com/

Table S6. Number of papers published each year for various algorithm types.

Paradigm/Year	**2015**	**2016**	**2017**	**2018**	**2019**	**2020**	**2021**	**2022**	**2023**	**2024**	**2025**
Reconstruction	10300	11100	11500	13000	14700	15200	19500	22700	25500	28200	13300
Forecasting	31000	33500	35100	40600	46300	52900	59700	64300	62300	60800	36100
Generative	8870	9640	9280	11500	13400	15600	19200	23200	24800	32000	16300
Similarity-based	12700	13600	14700	16400	18400	20700	24100	27500	30800	33900	17900
LLM-based	0	0	2	3	3	6	4	44	480	2600	2200
Tree-based	9570	10700	11400	12900	15000	17400	20800	24500	26900	24900	14500
Statistical	28700	30600	32800	37500	43000	49400	57100	57500	52100	46800	30900
Architecture/Year	**2015**	**2016**	**2017**	**2018**	**2019**	**2020**	**2021**	**2022**	**2023**	**2024**	**2025**
Class #1	98	126	137	193	284	366	539	882	1640	6230	3530
Class #2	436	462	1140	1400	1880	2530	4670	7970	13800	22100	13200
Class #3	847	921	902	1080	1130	1250	1510	1680	1800	2400	1300
Class #4	3630	4280	5100	7200	9880	13500	17900	22600	29100	31900	17000
Class #5	5390	6670	8480	11600	15600	19700	25400	33200	39800	42600	25300

B Algorithm Innovations

B.1 Classical Machine Learning

B.1.1 Forecasting Model. The Autoregressive Integrated Moving Average (**ARIMA**) is a forecasting method that predicts future time series data by analyzing historical data. In TSAD, it is assumed that anomalies represent a small portion of the data, allowing the prediction error to serve as the target error for the anomaly detection task.

B.1.2 Distance-based Model. K-Nearest Neighbors (**KNN**) [113] measures outlierness by assessing proximity to the k-th nearest neighbor, effectively detecting anomalies in TSAD. Dynamic Time Warping (**DTW**) [115] uses dynamic programming for non-linear alignment of two time series, determining their similarity. By comparing this similarity to an anomaly threshold, DTW can effectively identify anomalies.

B.1.3 Density-based Model. Wang and Qin [142] introduced a time series anomaly detection method that combines K-means with Particle Swarm Optimization (PSO). It uses Parallel PSO to find optimal initial clustering centers, improving K-means outcomes and addressing issues of cluster formation and local optima. Li et al. [83] enhanced Fuzzy C-Means clustering by modifying the Euclidean distance to adjust the significance of variables. This method employs PSO to optimize variable weights and quantifies anomaly levels by calculating reconstruction errors using optimal cluster centers and the membership function matrix. The Periodic Self-Organizing Map (PSOM) [177] improves traditional SOM by organizing neurons based on temporal periodicity, targeting input vectors to the winning neuron in each column, and detecting anomalies by assessing neighborhood differences. Local Outlier Factor (**LOF**) [16] detects local anomalies by assessing the ratio of local density to neighboring density as the anomaly score. Goldstein and Dengel [42] proposed an unsupervised histogram-based outlier detection algorithm (**HBOS**) that models the density of univariate features using histograms and calculates anomaly scores based on fixed or dynamic bin widths. The **ECOD** algorithm [85] independently calculates the Empirical Cumulative Distribution Function for each dimension to estimate tail probabilities, then aggregates these to derive a final anomaly score, making it widely used in TSAD.

Table S7. Summary of algorithmic innovations and their key references.

Algorithm Innovations	Subcategories	References
Training Method (Sec. 5.1)	AE-based Model	**EncDec-AD** [98], **MSCRED** [171], **UAE**[37], [54], **USAD** [8], **RE-ADTS** [7], **NPSR** [74]
	Contrastive Learning Model	**PatchAD** [186], **CARLA** [27], **SimAD** [185], **TimeAutoAD** [69], **USD** [90], **BroadCAE** [183]
	One Class-based Model	[99], **OC-TCN** [103], **COCA** [143], **COUTA** [156], **THOC** [120]
	Generative Adversarial Model	**FGANomaly** [33], [105], [169], **TadGAN** [39], **AnoFormer** [124], **BeatGAN** [93], **UAC-AD** [89]
	VAE Model	**LSTM-VAE** [109], [111], **InterFusion** [84], **MUTANT** [123], **FCVAE** [147]
Deep Learning Architecture (Sec. 5.2)	RNN Model	**OmniAnomaly** [130], **CAE-AD** [187], **MSDMM** [160], **LSGA** [100]
	GNN Model	[180], [31], **STADN** [38], **DGINet** [57], **DPGLAD** [47], **SG-Trans** [163]
	TCN Model	[9], [110], [114], [182], **TCN-AE** [136], **WaRTEm-AD** [1], [56], **TFAD** [172]
	State Space Model	**S-Mamba** [146], [86], [17], **DMamba** [18], **C-Mamba** [168], **Mambaformer** [159], **SGFM** [49], **TSAD-C** [51], **SGFM** [49], **TSAD-C** [51]
	Diffusion-based Model	**D3R** [140], **ImDiffusion** [19], [112], **DDMT** [162], **TimeADDM** [55], **DDTAD** [131]
	Transformer-based Model	**Anomaly Transformer** [157],**TransAnomaly** [173], **Cause-Former** [182] **TiSAT** [32], **TranAD** [137], **AnomalyBERT** [64], **DCdetector** [164], **D3TN** [139], [72]
	Prototype-based Model	**MEMTO** [128], **AMSL** [179], **AdaMemBLS** [184]
	Others	[24], [62], [158]
Classical Machine Learning (Appendix B.1)	-	**KNN**) [113], **DTW** [115], [142], [83], **PSOM** [177], **LOF** [16], **HBOS** [42], **ECOD** [85], **OCSVM** [118], **EB-OCSVM** [176], [12], [145], Isolation Forest [88]
Unified TSAD Model (Sec. 5.3)	-	**DADA** [121], **DACAD** [28], **ContexTDA** [75], **FS-ADAPT** [82], **VANDA** [48], **UNITS** [36]
LLM-based TSAD Model (Sec. 5.4)	Predictive LLM-based Model	**TS-Bert** [25], **GPT4TS** [188], **aLLM4TS** [13], **Timer** [94]
	Exploring LLM-Powered Model	**AnomalyLLM** [87], **LLMAD** [91], **SIGLLM** [6], **ParInfoGPT** [106]

B.1.4 Classification Model. The One-Class Support Vector Machine (OCSVM) [118] is a machine learning method that utilizes support vector machine principles for TSAD through a classification-based approach. Several variants of the OCSVM include: **EB-OCSVM** [176] utilizes OCSVM with an extended boundary, first identifying change points to establish a causal relationship graph among variables. It then trains an OCSVM model to define the classification boundary for normal values, extending this boundary to minimize noise errors. Upon detecting an anomaly, the method references the causal graph to identify the root cause. Bevilacqua and Tsaftaris [12] introduced a TSAD technique that integrates dictionary learning (DL) with OCSVM. The approach iteratively detects anomalies by first learning the normal time series pattern via DL, then applying an enhanced OCSVM model that assigns larger slack variables to data points with higher reconstruction errors from the DL phase. Wang et al. [145] improved the PSO algorithm by adding adaptive speed weighting and adaptive population splitting to optimize OCSVM parameters.

B.1.5 Tree-based Model. As noted by Liu et al. [88] and other researchers, the Isolation Forest model uses a tree-like structure to isolate anomalies. Given $\boldsymbol{t}$ as input, it randomly selects a feature and split point to partition data points into two subsets. This recursive process continues until each subset contains one data point or meets a stopping condition, resulting in multiple random split trees. The anomaly score for each data point is determined by its average path length across the trees, with lower scores indicating a higher likelihood of being anomalies. This approach is widely used in TSAD.

B.1.6 Statistical Method. The Z-Scores algorithm detects anomalies in time series data by calculating each point's deviation from the mean, normalized by the standard deviation. The 3-Sigma algorithm, assuming a normal distribution, identifies anomalies when a data point's deviation exceeds three standard deviations from the mean. The Interquartile Range (IQR) method uses quartiles to detect anomalies, computing the difference between the upper and lower quartiles and typically applying 1.5 times the IQR to identify outliers.

B.2 Training Method

B.2.1 Variational Autoencoder Model. The primary advantage of Variational Autoencoders (VAEs) in TSAD lies in their variational inference mechanism, which aligns generated samples with observed data by optimizing the parameters of the latent space distribution. This framework enhances the model's understanding of data structure, providing a robust foundation for anomaly detection by balancing reconstruction error and latent distribution consistency.

A notable implementation is **LSTM-VAE** [109], which combines LSTM with VAE to dynamically address task execution changes through state-based threshold selection.

In another approach, Pham et al. [111] utilize convolutional kernels of varying sizes to capture both long- and short-scale information, learning hidden space representations through variational inference. **InterFusion** [84] introduces a Hierarchical VAE that explores a low-dimensional two-view embedding strategy for time series data, employing MCMC-based imputation for improved accuracy in embeddings.

MUTANT [123] constructs feature maps for each time window, leveraging a graph neural network to learn variable embeddings, and incorporates a reconstruction model with attention mechanisms, LSTM, and a VAE module to assess variable importance. Lastly, **FCVAE** [147] integrates Global and Local Frequency Information Extraction modules, using frequency features as conditional factors within the CVAE framework and enhancing the capture of short-term trends through an attention mechanism. Together, these approaches illustrate the versatility and effectiveness of VAEs in addressing the complexities of anomaly detection in time series data.

B.3 Deep Learning Architecture

B.3.1 Recurrent Neural Network Model. Recurrent Neural Networks (RNNs) leverage their recurrent connections to capture long-term dependencies in time series data, making them highly effective for learning complex temporal patterns and detecting anomalies. Their ability to maintain temporal memory allows RNNs to handle sequences with significant temporal dynamics, which is crucial for TSAD.

For instance, **OmniAnomaly** [130] employs a stochastic RNN to explicitly explore temporal dependencies among random variables, utilizing Stochastic Gradient Variational Bayes and planar Normalizing Flows to estimate the posterior distribution of non-Gaussian variables. Similarly, **CAE-AD** [187], an LSTM-based algorithm, features a siamese structure that integrates self-attention for positional information extraction, alongside Discrete Fourier Transform-based data augmentation and Instance and Contextual Contrasting losses to capture local invariant features.

MSDMM [160] utilizes a GRU network to extract time series features, enhancing normal window reconstruction by retrieving global normal features through a memory module and constructing local memory from neighboring windows. **LSGA** [100] focuses on learning a sparse adjacency matrix between sensors via node embedding similarity and Gumbel-Softmax sampling, minimizing sensor relationship density while employing CNNs for short-term local pattern capture and Skip-GRU for long-term trend analysis.

To mitigate the issue of excessive temporal distribution overlap in RNN models across different time stages, Sun et al. [132] proposed the Ranking-Based Cross-Entropy Loss. This loss function jointly learns class features and the priority of temporal positions, thereby assisting the classifier in generating probability scores with clearer boundaries between distinct phases.

B.3.2 Graph Neural Network Model. Graph Neural Networks (GNNs) have emerged as powerful deep learning architectures designed specifically for graph-structured data, effectively capturing intricate relationships among nodes. In the realm of TSAD, GNNs are increasingly applied to graph-based time series data, encompassing dynamic social networks, traffic flow, and bioinformatics.

A notable approach is presented by Zhao et al. [180] with **MTAD-GAT**, which treats each univariate time series as an independent feature. This model leverages graph attention mechanisms to elucidate feature dependencies and temporal relationships, thereby enhancing interpretability. Additionally, it incorporates both prediction and reconstruction methodologies to improve the effectiveness of anomaly detection.

Similarly, Ding et al. [31] introduce **MST-GAT**, which employs multi-head attention alongside inter-modal and intra-modal learning to capture spatial features, coupled with convolutional networks for temporal feature extraction. This architecture integrates prediction and reconstruction techniques to deliver a joint anomaly scoring system.

In the work of [38], **STADN** constructs a sensor correlation matrix utilizing graph attention to extract spatiotemporal characteristics from time series data. Notably, it features an Error Refinement Module aimed at assisting users in establishing appropriate thresholds for anomaly detection.

Further extending the capabilities of GNNs, **DGINet** [57] combines a dynamic graph attention network with Informer to effectively capture correlations among sensor features. Meanwhile, **DPGLAD** [47] employs unidirectional graph structure learning to model causal relationships among sensors within a dynamic framework, utilizing Diffusion Convolution RNN with timestamp masking to capture both temporal and spatial features. **MagNet** [52] leverages wavelet transformations to categorize time series into multi-resolution time-frequency components. It incorporates an orthogonal kernel to restrict signals at varying frequencies, which helps in mitigating information redundancy. Furthermore, the model uses a frequency-layered GNN to capture dynamic spatial relationships, with the ultimate features being weighted and aggregated through average pooling.

Lastly, **SGTrans** [163] introduces a joint optimization approach involving a reconstruction module based on sparse autoencoders and a prediction module utilizing Graph Transformer, aimed at extracting spatiotemporal correlations across different time sequences.

B.3.3 Temporal Convolution Network Model. Temporal Convolution Networks (TCNs) are specialized deep learning architectures designed for efficient feature extraction in time series data, utilizing convolution operations to capture local patterns and temporal dependencies effectively.

The foundational work by Bai et al. [9] proposed the basic TCN, enhancing traditional CNN networks through one-dimensional convolutions to better learn temporal information. Building upon this, Park et al. [110] successfully applied TCN to various applications, including ECG anomaly detection and gesture recognition.

In a different approach, Ren et al. [114] introduced the Spectral Residual (SR) combined with a CNN, termed SR-CNN, which effectively captures saliency features within time series data. This model employs tail interpolation to actively identify anomalies, facilitating integration with online monitoring systems for rapid diagnostics.

DKN [154] constructs a dual-tower distillation network architecture, based on convolutional and attention networks. This design aims to extract both local and global features while enhancing information flow across different layers. It achieves dense mutual supervision between low-level and high-level semantic information, enabling DKN to uncover rich regularizations and relationships hidden within the data.

Thill et al. [136] introduced the **TCN-AE**, which incorporates multiple skip connections and dilated convolutions, improving the model's capacity to process long-distance dependencies. Their experimental validation demonstrated the advantages of skip connections and dimensionality reduction of feature maps in enhancing anomaly detection performance.

Moreover, Abilasha et al. [1] proposed **WaRTEm-AD**, which employs time warping operations for data augmentation in TSAD, including replication and interpolation warping techniques. They also introduced a twin TCN autoencoder structure aimed at learning warping-robust features. In a novel approach, Huang et al. [56] presented a multi-resolution self-supervised discriminative network for anomaly detection. This architecture utilizes multi-scale down-sampling and self-supervised modules, diverging from traditional reconstruction or prediction-based scoring methodologies. Their TCN block integrates Conv1d and max-pooling layers to enhance feature extraction.

Additionally, Zhang et al. [174] developed a multi-scale convolutional attention algorithm for anomaly detection. By employing one-dimensional convolution layers with varying receptive fields and an amplified attention mechanism, their model effectively emphasizes anomalies for improved identification. Lastly, Zhang et al. [172] introduced the **TFAD** architecture, which leverages time-frequency analysis to identify various component anomalies by utilizing decomposition details across multiple frequency domains.

C Evaluation Methods

C.1 Evaluation Metrics' Formulas

In Table S8, we provide a compilation of the formulas for the metrics introduced in the main text, along with necessary explanations and clarifications in the "Comment" column. Moreover, the R-based F1, TaPR-F1, and eTaPR F1 metrics share a common foundational logic, computed using the formula: $2 \cdot \frac{\text{Precision}_T \cdot \text{Recall}_T}{\text{Precision}_T + \text{Recall}_T}$. Specifically, all three metrics define Precision_T and Recall_T to weight interval predictions, integrating factors such as prediction accuracy, overlap, and positional fidelity. The final score adheres to a calculation paradigm analogous to the classical F1 metric, with the added nuance of incorporating anomaly persistence. Given the complexity inherent in their underlying principles and computational frameworks, we recommend consulting the original literature to avoid oversimplification and to maintain technical precision. Similarly, the definition of the more intricate VUS metric has not been introduced in this section.

Table S8. Overview of the formulas for evaluation metrics along with accompanying comments.

Metric	Formula	Comment
Precision	$\text{Precision} = \frac{\sum_{i=1}^{N} y_i \cdot \hat{y}_i}{\sum_{i=1}^{N} \hat{y}_i}$	Classical precision.
Recall	$\text{Recall} = \frac{\sum_{i=1}^{N} y_i \cdot \hat{y}_i}{\sum_{i=1}^{N} y_i}$	Classical recall.
F1-Score	$\text{F1} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$	Classical F1 score.
F1-PA [155]	$\hat{y}_i = \begin{cases} 1, & \text{if } \mathcal{F}_t > \delta \\ & \text{or } t \in S \text{ and } \exists_{t' \in S} \mathcal{F}_{t'} > \delta \\ 0, & \text{otherwise.} \end{cases}$	The F1-PA metric follows the same calculation framework as the classical F1-Score. However, F1-PA applies a post-processing adjustment to the predicted results $\hat{y}_i$. Here, S denotes an arbitrary anomaly segment, and $\mathcal{F}$ represents the anomaly score function.
Reduced F1 [125]	$\lvert S_i \rvert = \ln(\lvert S_i \rvert) + 1,$ $\text{Pre} = \frac{\sum_{i=1}^{\lvert S \rvert} S_i \cdot \hat{S}_i}{\sum_{i=1}^{\lvert S \rvert} \hat{S}_i}, \ \text{Rec} = \frac{\sum_{i=1}^{\lvert S \rvert} S_i \cdot \hat{S}_i}{\sum_{i=1}^{\lvert S \rvert} S_i},$ $\text{ReducedF1} = 2 \cdot \frac{\text{Pre} \cdot \text{Rec}}{\text{Pre} + \text{Rec}}$	Reduced F1 also employs a point adjustment strategy, but its improvement lies in adjusting the length of anomaly segments. For any anomaly segment S_i, after point adjustment, the anomaly length is post-processed as shown in the first formula. The F1 score is then calculated on an event-by-event basis to mitigate the bias introduced by point adjustments.
PA%K [73]	$\hat{y}_i = \begin{cases} 1, & \text{if } \mathcal{F}_t > \delta \text{ or} \\ & t \in S \text{ and } \frac{\lvert \{t' \mid t' \in S, \mathcal{F}_{t'} > \delta\} \rvert}{\lvert S \rvert} \geq \text{K} \\ 0, & \text{otherwise} \end{cases}$	PA%K is an improvement over F1-PA, modifying the original point adjustment strategy. In the traditional approach, an entire anomaly segment S was considered successfully predicted if any single point within the segment was correctly identified. By contrast, PA%K requires that the proportion (e.i. $\frac{\lvert \{t' \mid t' \in S, \mathcal{F}_{t'} > \delta\} \rvert}{\lvert S \rvert}$) of correctly predicted anomaly points within a segment exceeds $K\%$ (where K is a pre-defined value ranging from 0 to 100) to trigger post-processing and mark the segment as successfully predicted. When K varies across 0 to 100, the area under the PA%K curve can be computed as a new metric, denoted by $F1_{\text{PA\%K}}$.
Affiliation F1 [58]	$S = \{j \in [1, n]; \text{pred} \cap I_j \neq \emptyset\},$ $\text{Aff-Pre} = \frac{1}{\lvert S \rvert} \sum_{j \in S} P_{\text{precision}_j},$ $\text{Aff-Rec} = \frac{1}{n} \sum_{j=1}^{n} P_{\text{recall}_j}.$ $\text{Aff-F1} = 2 \cdot \frac{\text{Aff-Pre} \cdot \text{Aff-Rec}}{\text{Aff-Pre} + \text{Aff-Rec}}.$	In the Aff-F1 metric, the predicted results are partitioned into n intervals, where I_j denotes the j-th zone of affiliation of the ground truth event. Aff-Pre and Aff-Rec denote Affiliation Precision and Affiliation Recall, respectively.
N&U Affs [185]	$\text{UAff-Pre} = \frac{\text{Aff-Pre} - \text{Aff-Pre}_{bias}}{1 - \text{Aff-Pre}_{bias}},$ $\text{UAff-F1} = \frac{2 \cdot \lvert \text{UAff-Pre} \rvert \cdot \text{Aff-Rec}}{\lvert \text{UAff-Pre} \rvert + \text{Aff-Rec}} \cdot (-1)^{\lvert \text{UAff-Pre} < 0 \rvert}.$ $\text{NAff-Pre} = 2(\text{Aff-Pre} - 0.5),$ $\text{NAff-F1} = \frac{2 \cdot \lvert \text{NAff-Pre} \rvert \cdot \text{Aff-Rec}}{\lvert \text{NAff-Pre} \rvert + \text{Aff-Rec}} \cdot (-1)^{\lvert \text{NAff-Pre} < 0 \rvert}.$	NAff-F1 and UAff-F1 denote Normalized Affiliation F1 and Unbiased Affiliation F1, respectively. Here, $\text{Aff-Pre}_{\text{bias}}$ represents the dataset-related precision bias estimated by the algorithm.
PATE [40]	$\text{Precision}_{e,d}(\theta) = \frac{\sum_{t=1}^{N} w^{\text{TP}}(t)}{\sum_{t=1}^{T} w^{\text{TP}}(t) + \sum_{t=1}^{N} w^{\text{FP}}(t)}$ $\text{Recall}_{e,d}(\theta) = \frac{\sum_{t=1}^{N} w^{\text{TP}}(t)}{\sum_{t=1}^{N} w^{\text{TP}}(t) + \sum_{t=1}^{N} w^{\text{FN}}(t)},$ $\text{PATE} = \frac{1}{\lvert E \rvert \times \lvert D \rvert} \sum_{e \in E, d \in D} \text{AUC} - \text{PR}_{e,d}$	In PATE, the weights for true positive rate, false positive rate, and false negative rate are defined as $w^{\text{TP}}(t)$, $w^{\text{FP}}(t)$, and $w^{\text{FN}}(t)$, respectively. The overall PATE score is determined by averaging the computed AUC-PRs across all combinations of e and d, where $\lvert D \rvert$ and $\lvert E \rvert$ denote the number of distinct values for d and e within their respective sets.
AUC-ROC	$\int_0^1 \text{TPR}(\text{FPR}) \, d\text{FPR}$	It is obtained by calculating the area under the curve of the True Positive Rate (TPR) and False Positive Rate (FPR).
AUC-PR	$\int_0^1 \text{Precision}(\text{Recall}) \, d\text{Recall}$	Here, Precision and Recall respectively denote the precision curve and the recall curve.

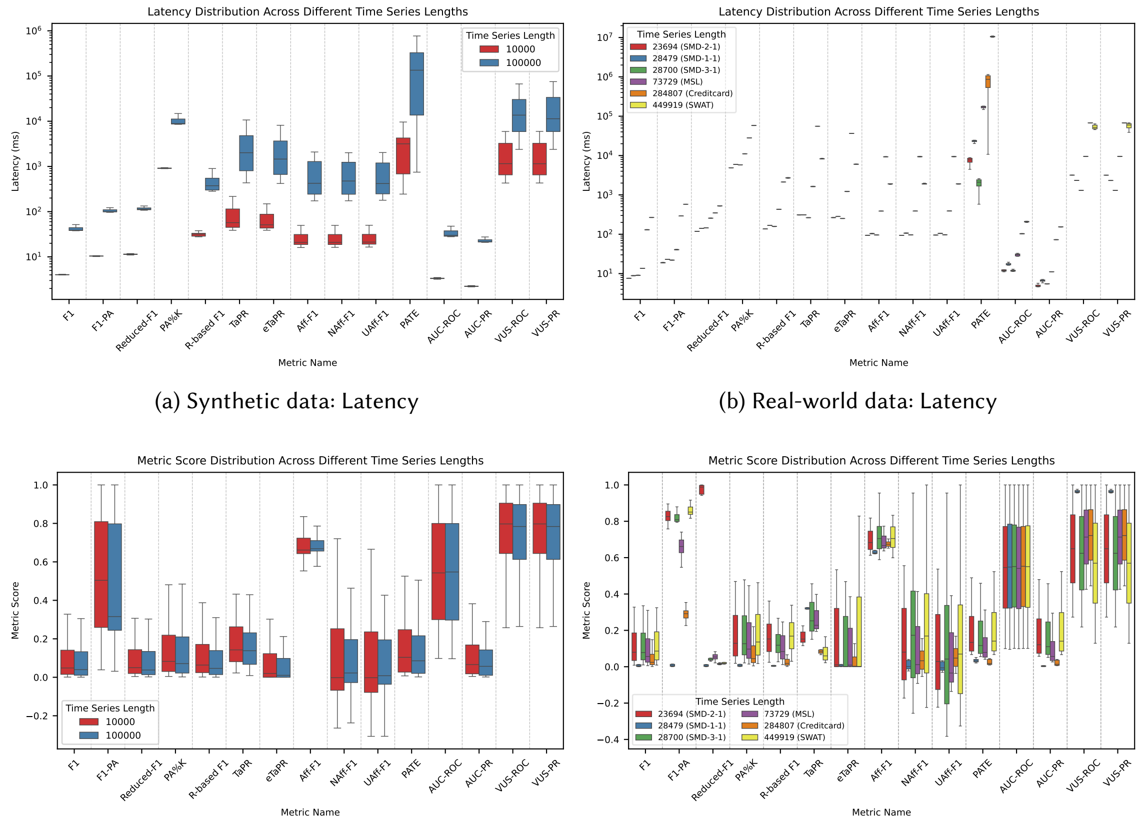
C.2 Practical Recommendations for Metric Selection

Table S9. Time series characteristics of synthetic and real data.

Dataset Name	TS Length	Segments	Max Seq Length	Min Seq Length
100k-20seg-50L	100000	20	60	40
100k-200seg-50L	100000	200	60	40
100k-20seg-50H	100000	20	99	1
100k-200seg-50H	100000	200	99	1
100k-50seg-20L	100000	50	30	10
100k-500seg-20L	100000	500	30	10
100k-50seg-20H	100000	50	39	1
100k-500seg-20H	100000	500	39	1
100k-10seg-100L	100000	10	110	90
100k-100seg-100L	100000	100	110	110
100k-10seg-100H	100000	10	199	1
100k-100seg-100H	100000	100	199	1
100k-2seg-500L	100000	2	550	450
100k-20seg-500L	100000	20	550	450
100k-2seg-500H	100000	2	999	1
100k-20seg-500H	100000	20	999	1
10k-2seg-50L	10000	2	60	40
10k-20seg-50L	10000	20	60	40
10k-2seg-50H	10000	2	99	1
10k-20seg-50H	10000	20	99	1
10k-5seg-20L	10000	5	30	10
10k-50seg-20L	10000	50	30	10
10k-5seg-20H	10000	5	39	1
10k-50seg-20H	10000	50	39	1
10k-1seg-100L	10000	1	110	90
10k-10seg-100L	10000	10	110	110
10k-1seg-100H	10000	1	199	1
10k-10seg-100H	10000	10	199	1
10k-2seg-500L	10000	2	550	450
10k-2seg-500H	10000	2	999	1
MSL	73729	36	1141	11
Creditcard	284807	465	5	1
SWAT	449919	35	35900	101
SMD-1-1	28479	8	721	2
SMD-2-1	23694	13	452	8
SMD-3-1	28700	4	131	21

The selection of an appropriate evaluation metric is contingent upon the specific objectives of the analysis and the inherent characteristics of the time series data and anomaly detection task. Drawing from our experimental analysis, the following practical strategies and recommendations are provided:

(1) **Consider Latency Requirements:** When evaluation speed is a paramount concern, particularly for very long time series (> 100k) or datasets containing numerous anomaly segments (> 50), **PATE should be avoided**. In



(a) Synthetic data: Latency

(b) Real-world data: Latency

(c) Synthetic data: Performance

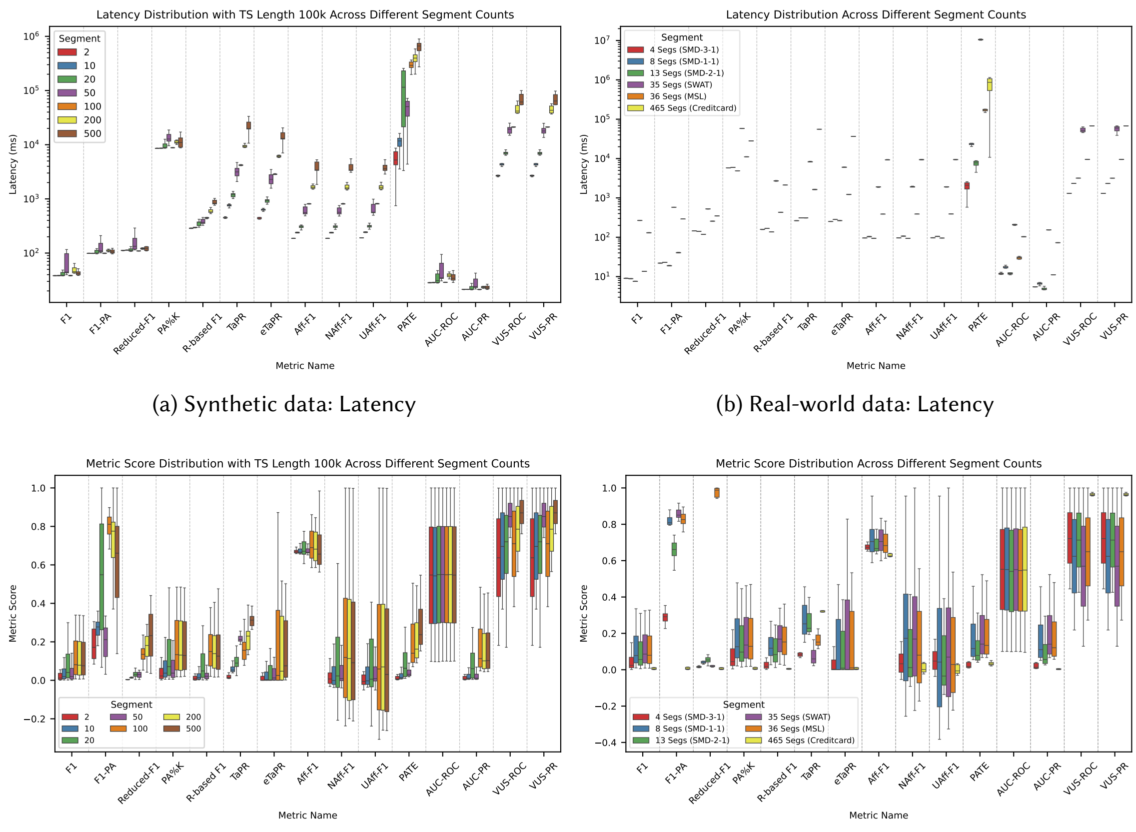
(d) Real-world data: Performance

Fig. S8. Performance variation in latency and metrics across different **time series lengths** for synthetic and real-world data.

such scenarios, point-based metrics, or interval-based metrics less affected by segment count such as F1-PA, Reduced-F1, and PA%K, are preferable for achieving lower latency. It is important to note that the latency of many interval-based metrics tends to increase with the number of anomaly segments.

(2) **Understand Anomaly Characteristics:**
   - For scenarios involving **short anomaly segments or low variance in segment length**, metrics such as F1, F1-PA, PA%K, Reduced-F1, TaPR, PATE, and AUC-PR may be more appropriate, given their reduced bias under these conditions.
   - Conversely, for **long anomaly segments or high variance in segment length**, NAff-F1, UAff-F1, VUS-ROC, and VUS-PR generally demonstrate superior performance with lower inherent bias.

(3) **Evaluate Discriminability and Robustness:**
   - To effectively **discriminate between models that are not perfectly accurate** (e.g., exhibiting 10%–90% accuracy), caution is advised when using Reduced-F1, PATE, and AUC-PR, as these metrics have demonstrated poor discriminability for such models.



(a) Synthetic data: Latency

(b) Real-world data: Latency

(c) Synthetic data: Performance

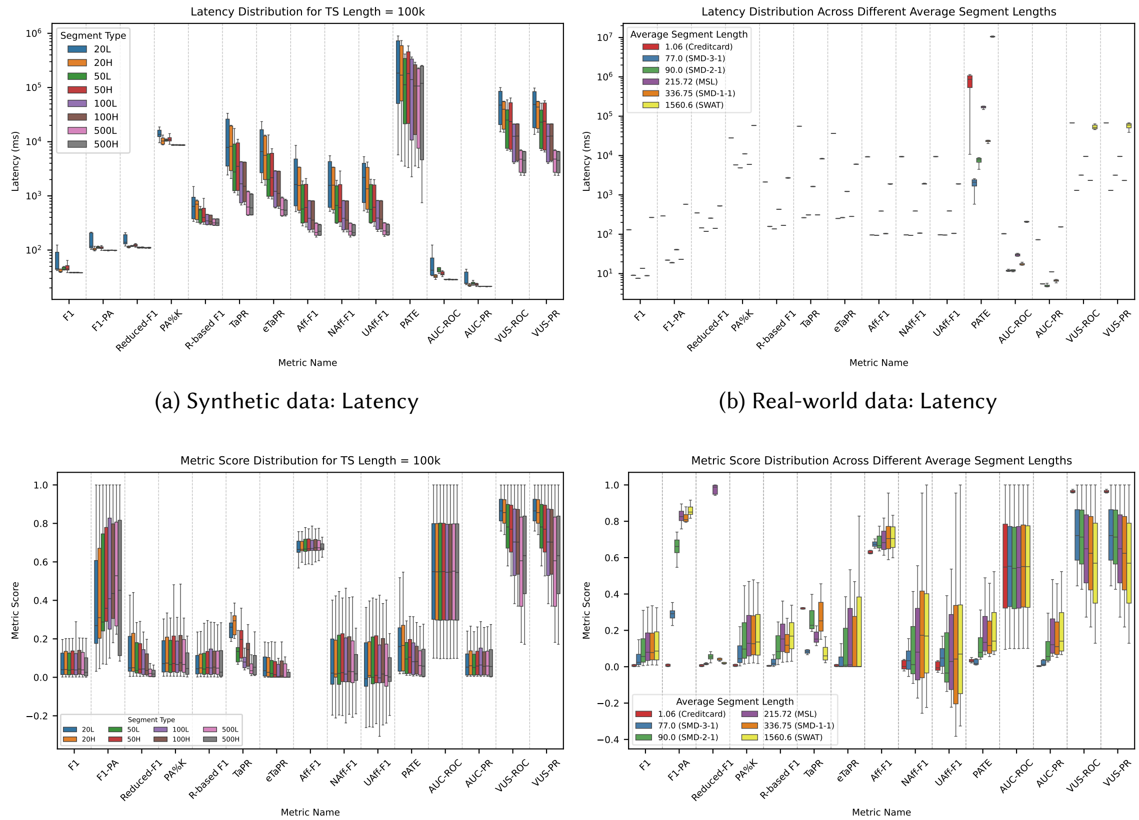
(d) Real-world data: Performance

Fig. S9. Performance variation in latency and metrics across the **number of segments** for synthetic and real-world data.

- F1 and F1-PA exhibit sensitivity to noise and variations in threshold. Consequently, **F1-PA should not be employed in isolation** for real-world evaluations due to its variable bias baseline and high score variance. Its use should be complemented by other, more stable metrics.
- It is noteworthy that the inability of R-based F1 and TaPR to achieve theoretical maximums may result in a consistent underestimation of performance, a factor of significance when absolute scores are critical.

(4) **Threshold Dependency:** For **threshold-independent evaluation**, VUS metrics (VUS-ROC, VUS-PR) represent robust options. Nevertheless, it is imperative to **standardize the event buffer size** across all comparative analyses to preclude the introduction of artificial bias.

(5) **Interpretability of Metric:** In practical applications where an intuitive understanding of model performance is required, metrics that rely on point adjustment, including F1-PA, PA%K, and TaPR, are generally not recommended. Metrics such as F1, AUC-ROC, and AUC-PR provide a direct interpretation of the model's point-wise detection performance. Conversely, eTaPR, Aff-F1, NAff-F1, UAff-F1, PATE, VUS-ROC, and VUS-PR offer a



(a) Synthetic data: Latency (b) Real-world data: Latency

(c) Synthetic data: Performance (d) Real-world data: Performance

Fig. S10. Performance variation in latency and metrics across different segment lengths for synthetic and real-world data.

direct assessment of the model's interval anomaly detection capabilities. A detailed explanation of all metrics can be found in Table 3.

(6) **Utilize Multiple Metrics for a Holistic View:** Given the inherent biases and sensitivities of individual metrics, it is often considered best practice to **report results utilizing a combination of metrics**. This approach facilitates a more comprehensive and robust evaluation of model performance by mitigating the limitations of any single metric. For instance, pairing a threshold-dependent metric with a threshold-independent one, or a segment-based metric with a point-based one, can provide a more balanced perspective.

By carefully considering these factors, researchers can make an informed decision regarding the most suitable metrics for their specific time series anomaly detection evaluation requirements.

D Applications

In this part, we further elaborate on the applications of TSAD algorithms. Specifically, we organize the overview from three aspects: (1) the application fields of the algorithms, (2) the regularly used datasets in academic research, (3) and the algorithm libraries. The objective of this part is to provide a more particular supplement to the application domains of the algorithms. At the same time, it intends to aid beginners in the field of TSAD by enabling them to practice more quickly.

D.1 Application's Field

In this section, we review how TSAD has been applied in several classic domains and summarize, in brief, the algorithmic requirements specific to each field.

Business KPI Anomaly Detection: Laptev et al. [78] introduces EGADS, a scalable framework for automated anomaly detection, effectively implemented at Yahoo to monitor millions of time series data. This application has revolutionized the monitoring of system metrics and business KPIs, facilitating efficient detection of infrastructure and product issues. By open-sourcing the framework and datasets, the paper encourages community contributions that capture customer experiences on Yahoo sites, thereby enhancing operational decision-making and strategic planning. **This field requires TSAD models** that can handle large volumes of data and have lower real-time processing demands. The domain primarily focuses on monitoring system metrics and business KPIs, dealing with millions of time series. Its emphasis is on efficiently processing vast volumes of data to support operational decisions and strategic planning, rather than on delivering real-time responses.

Power Failure Anomaly Detection: Beattie et al. [10] presents a robust and interpretable data-driven method for anomaly detection in power electronic devices, facilitating real-time detection of anomalies in the data streams of power electronic converters, essential for maintaining complex production systems. This approach is also well-suited for integration with IoT, enhancing the smart grid's communication, control, and computing technologies. **This field requires TSAD models** that are robust, interpretable, and meet real-time processing demands. Because real-time anomaly detection in power-electronic data streams is essential to the stable operation of complex production systems, it enables prompt equipment maintenance; robustness and interpretability further help engineers understand the root causes of anomalies and take effective actions. The growing convergence of this field with the Internet of Things likewise heightens the demand for low-latency detection.

Vehicle Trajectory Anomaly Detection: DiffTAD [79] is the first model to apply diffusion models for vehicle trajectory anomaly detection, framing the process as a transition from noise to normalcy. The model incrementally adds noise to vehicle trajectories until they resemble Gaussian noise, then learns the inverse process to reconstruct the original trajectories. DiffTAD can quickly identify and evaluate anomalous behavior in specific areas, which is vital for intelligent traffic monitoring, aiding in the prevention of traffic accidents and illegal activities. By proactively detecting abnormal behavior of surrounding vehicles, DiffTAD helps autonomous vehicles choose safer driving paths, enhancing the overall safety and reliability of the system. **This field requires TSAD models** that can simultaneously model spatiotemporal information, with a focus on the influence of spatial relationships on anomalies. Vehicle trajectory data exhibit changes over time and shifts in spatial location. Spatial relationships such as inter-vehicle distance and relative position are critical for detecting anomalies, so the model must jointly model spatial and temporal information.

Physiological Signal Anomaly Detection: ECGGAN [141] is an interpretable model for ECG anomaly detection that segments ECG data into individual heartbeat instances using the maximum peak R-wave, simplifying reconstruction. It captures distinctive features across different leads to train the reconstruction model and learn normal ECG patterns. Anomalies are detected based on reconstruction error, with larger errors indicating a higher likelihood of abnormality. This method enhances interpretability and aids accurate clinical diagnosis, enabling automatic monitoring of ECG data for anomalies while providing relevant explanations to safeguard patient health. **This field requires TSAD models**

with high interpretability and a strong ability to learn complex relationships between channels. Anomaly detection in physiological signals directly affects patient safety, so clinicians must understand the basis of each alert; the model must therefore be interpretable. Such signals typically contain multiple leads whose complex inter-channel relationships must be accurately learned to ensure reliable detection.

Industrial Predictive Maintenance: Hsieh et al. [53] proposed an unsupervised TSAD method for predictive maintenance in Smart Manufacturing, combining LSTM and Autoencoder models. The LSTM captures temporal dependencies in multivariate data, while the Autoencoder learns normal patterns. After training for low reconstruction error, the model processes online data for real-time anomaly detection, triggering alerts if most predicted labels are anomalies. To mitigate label scarcity and extended training times, the algorithm uses transfer learning, facilitating predictive maintenance that reduces maintenance frequency, saves costs, and ensures normal operations. **This field requires TSAD models** that prioritize high real-time performance, the ability to process time-series data online, and algorithms with low spatial and temporal complexity. In industrial settings, real-time anomaly detection enables prompt maintenance actions that prevent minor faults from escalating. Online processing capabilities ensure the system can handle continuously arriving data streams, while low time- and space-complexity guarantee efficient operation within the computational resource constraints of industrial equipment.

D.2 Datasets

Table S10 compiles the most commonly used datasets in state-of-the-art methods, along with advanced synthetic datasets such as GutenTAG and TSB-UAD. We have summarized the links for fully open access and download. Additionally, in this table, we have documented the domain/application of each dataset, as well as its characteristics. For instance, #Subset indicates the number of subsets within a dataset, which often vary in time series length, collection location, or individual subject. Avg TS Len represents the average length of time series in each subset, Dim denotes the dimensionality of the time series data (*i.e.*, the number of features), and Anomaly Ratio indicates the average proportion of anomalies within the dataset. All these datasets are labeled. Furthermore, GutenTAG is a synthetic data generation toolkit that can produce diverse data subsets with customizable properties such as the number of channels and anomaly ratio. Therefore, its intrinsic dataset characteristics are not documented. These datasets cover common scenarios in TSAD, including server operation monitoring, human health tracking, and specific anomaly detection in industrial and commercial settings.

Table S10. Time series datasets from different domains.

Name	Type	Paper	Public Access Link	Domain/Application	#Subset	Avg TS Len	Dim	Anomaly Ratio (%)
IOPS	real	[114]	https://github.com/iopsai/iops	Business KPI	58	72792.3	1	1.3
Yahoo	real	-	https://webscope.sandbox.yahoo.com/catalog.php?datatype=s&did=70	Business KPI	367	1560.2	1	2.5
UCR	real	[29]	https://wu.renjie.im/research/anomaly-benchmarks-are-flawed/#ucr-timeseries-anomaly-archive	Health tracking	250	67818.7	1	0.6
ASD	real	[84]	https://github.com/zhhlee/InterFusion	Application server	12	6423.79	19	4.61
SMD	real	[130]	https://github.com/NetManAIOps/OmniAnomaly	Server machine	28	25466.4	38	3.8
WaQ	real	-	https://www.spotseven.de/gecco/gecco-challenge	Water quality	1	230172	9	2.02
MSL	real	[59]	https://github.com/khundman/telemanom	Sensor	27	3119.4	55	5.1
SMAP	real	[59]	https://github.com/khundman/telemanom	Sensor	54	7855.9	25	2.9
WADI	real	[4]	https://itrust.sutd.edu.sg/itrust-labs_datasets/	Water distribution	241921	34561	123	5.74
SWaT	real	[41]	https://itrust.sutd.edu.sg/itrust-labs_datasets	Secure water treatment	4	207457.5	59	12.7
SKAB	real	-	https://www.kaggle.com/dsv/1693952	Water circulation system	34	1200	8	35.48
GAIA	real	-	https://github.com/CloudWise-OpenSource/GAIA-DataSet	Log detection	279	10156.41	1	0.87
Exathlon	real	[63]	https://github.com/exathlonbenchmark/exathlon	Spark application monitoring	32	44075.8	1	11.0
Creditcard	real	-	https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud	Financial fraud detection	1	284807.0	29	0.2
CoMuT	synthetic	[151]	https://hpi.de/naumann/s/comut	Synthetic data	180	10000	9	0.93
NAB	real/synthetic	[3]	https://www.kaggle.com/datasets/boltzmannbrain/nab	Streaming media	58	5099.7	1	10.6
GutenTAG	synthetic	[150]	https://github.com/TimeEval/gutentag	Synthetic data tool	-	-	-	-
TSB-UAD	real/synthetic	[108]	https://github.com/TheDatumOrg/TSB-UAD	Dataset collection	1125	91694.87	1	5.95

D.3 Algorithm's Library

Table S11 compiles the most popular state-of-the-art methods for TSAD, categorized into various paradigms: Reconstruction (Rec), Forecasting (Fcst), Generative (Gen), Similarity-based (Sim), Classification (Cls), and LLM. Methods can belong to multiple paradigms but are listed under the one with the highest relevance.

Table S11. Code sources of different state-of-the-art algorithms.

Algorithm	Paradigm	Category	Public Access Link
AMSL	Rec	Prot	https://github.com/zhangyuxin621/AMSL
MEMTO	Rec	Prot	https://github.com/gunny97/MEMTO
PatchAD	Sim	Const	https://github.com/EmorZz1G/PatchAD
SimAD	Sim	Const	https://github.com/EmorZz1G/SimAD
DCdetector	Sim	Trans	https://github.com/DAMO-DI-ML/KDD2023-DCdetector
USD	Rec	Const	https://github.com/zangzelin/code_USD
TS-TCC	Rec	Const	https://github.com/emadeldeen24/TS-TCC
COUTA	Cls	OC	https://github.com/xuhongzuo/couta
COCO	Cls	OC	https://github.com/ruiking04/COCA
OCAN	Cls	OC	https://github.com/PanpanZheng/OCAN
FGANomaly	Gen	GAN	https://github.com/sxxmason/FGANomaly
BeatGAN	Gen	GAN	https://github.com/hi-bingo/BeatGAN
UAC-AD	Gen	GAN	https://github.com/lhysgithub/UAC-AD
MSCRED	Rec	AE	https://github.com/7fantasysz/MSCRED
UAE	Rec	AE	https://github.com/astha-chem/mvts-ano-eval
USAD	Rec	AE	https://github.com/manigalati/usad
NPSR	Rec	AE	https://github.com/andrewlai61616/NPSR
InterFusion	Rec	VAE	https://github.com/zhhlee/InterFusion
MUTANT	Rec	VAE	https://github.com/Coac-syf/MUTANT
FCVAE	Rec	VAE	https://github.com/CSTCloudOps/FCVAE
OmniAnomaly	Rec	RNN	https://github.com/NetManAIOps/OmniAnomaly
MTAD-GAT	Rec	GNN	https://github.com/ML4ITS/mtad-gat-pytorch
TFAD	Rec	TCN	https://github.com/DAMO-DI-ML/CIKM22-TFAD
AnomalyTrans	Rec	Trans	https://github.com/thuml/Anomaly-Transformer
AnomalyBERT	Rec	Trans	https://github.com/Jhryu30/AnomalyBERT
CauseFormer	Rec	Trans	https://github.com/zhongguoxiang/CauseFormer
S-Mamba	Rec	SSM	https://github.com/wzhwzhwzh0921/S-D-Mamba
Mambaformer	Rec	SSM	https://github.com/XiongxiaoXu/Mambaformer-in-Time-Series
D3R	Rec	Diff	https://github.com/ForestsKing/D3R
ImDiffusion	Rec	Diff	https://github.com/17000cyh/IMDiffusion
Diffussion	Rec	Diff	https://github.com/fbrad/DiffusionAE
DDMT	Rec	Diff	https://github.com/yangchaocheng/DMDDT
TimeADDM	Rec	Diff	https://github.com/Hurongyao/TIMEADDM
GANF	Rec	Other	https://github.com/EnyanDai/GANF
UniTS	Rec	LLM	https://github.com/mims-harvard/UniTS
Timer	Fcst	LLM	https://github.com/thuml/Large-Time-Series-Model
GPT4TS	Fcst	LLM	https://github.com/DAMO-DI-ML/One_Fits_All
SIGLLM	Fcst	LLM	https://github.com/sintel-dev/sigllm
aLLM4TS	Fcst	LLM	https://github.com/yxbian23/aLLM4TS
AnomalyLLM	LLM	LLM	https://github.com/fly-orange/AnomalyLLM